

Chemito ITS Protocol

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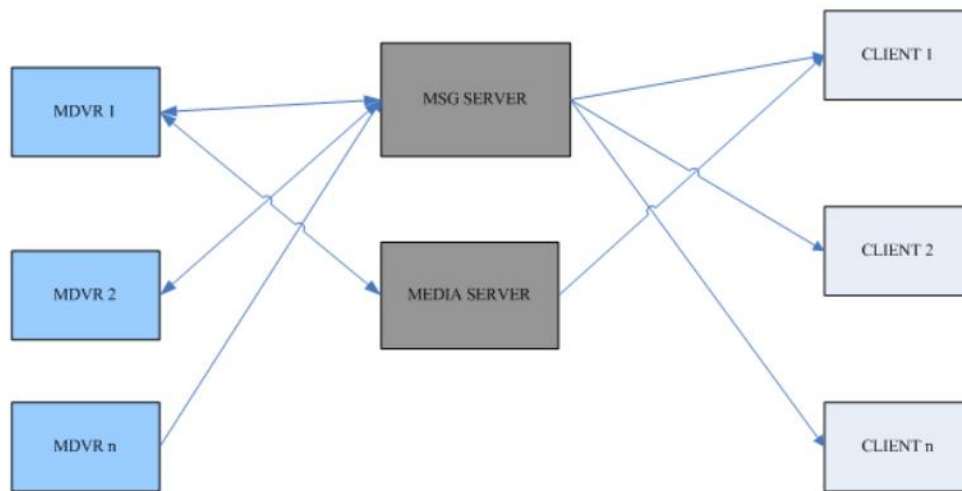
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Overview

The Description Mode



The graph above is the usage scenario of the forward mode: the device automatically establishes TCP persistent connection with the server after the boot, which effectively solves the problem of random assignment of the device's IP address.



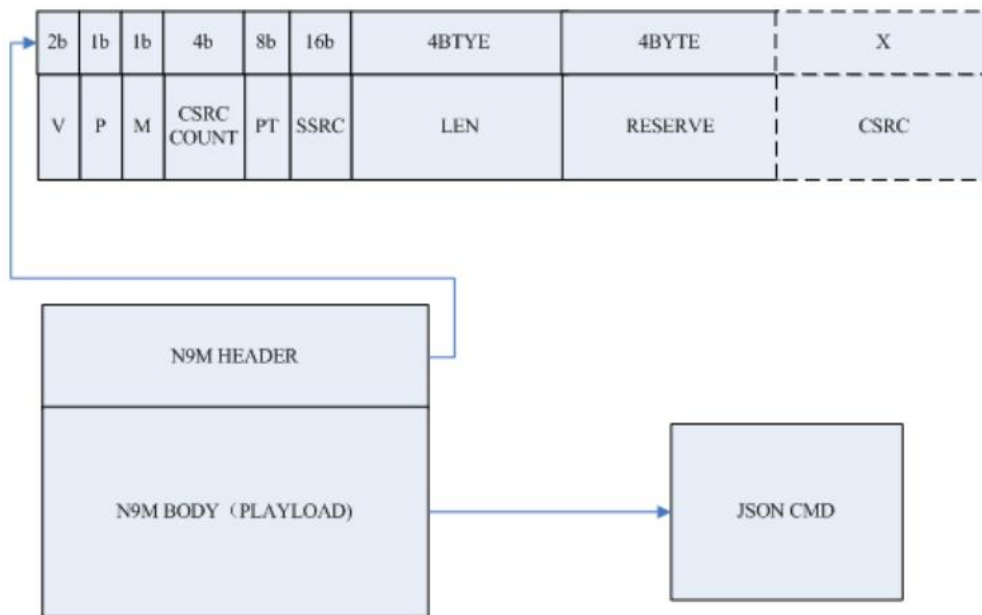
Establish TCP persistent connection (the server address can be set by the server configuration interface on the device interface) with the message server after the device boot. Then, messages like the server status, alarm and control command of the server are sent through this connection.

For media connection, the server requests the media command to send message to the device through the message connection. The message includes media server address and the device automatically connects to this media address for media transmission.

Protocol Encapsulation

All commands include N9M header and N9M body. For more details about N9M header, please refer to: protocol encapsulation . Body is usually a JSON string.

The command structure is shown below:



Some CSRC determines the length of concrete data according to CSRC COUNT. Generally, CSRC doesn't exist when CSRC COUNT is 0.

In C programming language, the HEAD part in the N9M protocol can be shown in the following structure:

```
typedef struct _packhead
{
    unsigned int V :2;//it is 2 bits
    unsigned int P :1;//it is 1 bit
    unsigned int M :1;//it is 1 bit
    unsigned int CSRC :4;//it is 4 bits
    unsigned int PT :8;//it is 8 bits
    unsigned int SSRC :16;//it is 16 bits
    unsigned int counts ;//it must use ntohl or htonl
    unsigned int reserve;
}__attribute__((packed)) _head;
```

Example Of Connection Process

Step1: the server listens to the port and configures the server address through the device UI

Step2: the device logs in the remote server

The device sends login command to the server (the following command is analog data and the value of JSON field needs to be filled in as needed) after establishing connection with the server

```
{
  "MODULE": "CERTIFICATE",
  "OPERATION": "CONNECT",
  "PARAMETER": {
    "AUTOCAR": "",
    "AUTONO": "0",
    "CARNUM": "",
    "CHANNEL": 5,
    "CID": 0,
    "CNAME": "CUSTOMER_04.1245",
    "DEVCLASS": 4,
    "DEVNAME": "MDVR",
    "DEVTYPE": 1,
    "DLIP": [{"LIP": "0.0.0.0", "MT": "eth0"}],
    "DLP": [80, 9006],
    "DSNO": "00600052B8",
    "EID": "",
    "LINENO": "",
    "MAC": [{"IMAC": "00:00:00:00:00:00", "MT": "eth0"}],
    "MODE": 1,
    "MTYPE": 18,
    "NET": 0,
    "PRO": "1.0.4",
    "STYPE": 16,
    "TSE": 0,
    "UNAME": "",
    "UNO": ""
  },
  "SESSION": "45f12f35-6fb5-49b6-b932-b1d6e20b0416"
}
```

The server responds to this login result:

```
{
  "MODULE": "CERTIFICATE",
  "OPERATION": "CONNECT",
  "RESPONSE": {
    "DEVTYPE": 1,
    "ERRORCAUSE": "",
    "ERRORCODE": 0,
    "MASKCMD": 1,
    "PRO": "1.0.4",
    "VCODE": ""
  },
  "SESSION": "45f12f35-6fb5-49b6-b932-b1d6e20b0416"
}
```

Note: SESSION of all subsequent commands is the SESSION used for the first login

Step3: the server sets parameters to the device. For more details about the parameter content, please refer to parameter documents and take new GPS data format for example (the following command won't be sent, and GPS is still uploaded in a JSON format, which consumes a lot of data traffic)

```
{
  "MODULE": "CONFIGMODEL",
  "OPERATION": "SET",
  "PARAMETER": {
    "MDVR": {
      "KEYS": {"GV": 1},
      "PGDSM": {"PGPS": {"EN": 1}},
      "PIS": {"PC041245T": {"GU": {"EN": 1, "TT": 5}}},
      "PSI": {"CG": {"UEM": 0}}
    }
  },
  "SESSION": "45f12f35-6fb5-49b6-b932-b1d6e20b0416"
}
```

The result of setting device response parameters:

```
{
  "MODULE": "CONFIGMODEL",
  "OPERATION": "SET",
  "RESPONSE": {
    "ERRORCAUSE": "SUCCESS",
    "ERRORCODE": 0
  },
  "SESSION": "45f12f35-6fb5-49b6-b932-b1d6e20b0416"
}
```

Step4: keep heartbeat. TCP persistent connection established between the device and server determines whether the link is normal according to the heartbeat package. The command is sent to the server from the device.

```
{
  "MODULE": "CERTIFICATE",
  "OPERATION": "KEEPALIVE",
  "SESSION": "45f12f35-6fb5-49b6-b932-b1d6e20b0416"
}
```

Heartbeat response:

```
{
  "MODULE": "CERTIFICATE",
  "OPERATION": "KEEPALIVE",
  "SESSION": "45f12f35-6fb5-49b6-b932-b1d6e20b0416"
}
```

Protocol Encapsulation

Encapsulation Header

All messages and media data adopt unified encapsulation headers

1-2bit	3bit	4bit	5-8bit	9-16bit	17-32bit
V	P	M	CSRC COUNT	PAYLOAD TYPE	SSRC
PAYLOAD LEN					
RESERVE					
CSRC(VARIABLE 0-15 ITEMS 32BITS EACH)					
PAYLOAD					

V : version number. The protocol version is 1

P : whether to encrypt. 0-don't encrypt and 1-encrypt. If data content also adopts the compress mode, then compress it before encryption.

M : whether to compress. 0-don't compress and 1-use gzip

CSRC COUNT : the range is 0-16. The current value is 8 and waits for expansion. The 1st to 3rd 32BITS in CSRC indicate source address and 4th to 7th 32BITS indicate target address. Each address has 16 bytes. The first 6 bytes is MAC address and the remaining 10 bytes are the unique flag generated by each system. The server can differentiate to which client the information is forwarded via target address. The same client can determine to which business connection the data belongs via the address.

PAYLOAD TYPE : The field is used to differentiate different command types. For more details, please refer to the table below

SSRC : different PAYLOAD has different implications. For more information, please refer to the table below

PAYLOAD LEN : the length of protocol content

RESERVE : reserve. The current value is 0 and waits for future expansion

CSRC : it varies according to commands. For details, please refer to the description of command. CSRC COUNT is 0 if there is no description, indicating CSRC doesn't exist

PAYLOAD : protocol data

Payload_Type_Table

PayloadType	Description	SSRC
0	indicates command data. Status and control commands all use this value. The data after this value is shown as JSON	reserved. No meaning yet
2	real-time video data. The direct connection mode. Different video streams have different PT values and forward modes. For various video stream types, PT values are different	indicates the channel number of video stream, which starts from 0.
3	recording data used for downloading recording files	reserved. No meaning
4	video playback data	indicates the channel number of playback data, which starts from 0
6	captured photos	reserved. No meaning
10	parameter import	reserved. No meaning
11	parameter export	the same as SSRC in request command. It is used to differentiate data source in media links.
15	transmission sub-stream	indicates the channel number of video stream, which starts from 0.
16	recording sub-stream	indicates the channel number of video stream, which starts from 0.
17	black box data	reserved. No meaning
22	special command	such as GPS, refer to Special Report
30	streamax maintain data	not opensourced currently
other	not used	reserved. No meaning

Message Format and Defination

Message format adopts Json mode for encapsulation. Message encapsulation should correspond to equipment description files. Each piece of message is shown in a string and uses big endian for storage.

JSON_Format

field	type	description
MODULE	string	indicates the command type. For more details, please refer to the follow-up business introduction
SESSION	string	identification connection. Identification connection is generated by the N9M host no matter it is directly connected or forwarded. When the client or server sends command to the N9M mutually, identification needs to be sent back
OPERATION	object	filling-in of the request command, indicating the operation type. For different MODULE, the definition of value varies. For more information, please refer to the follow-up business introduction
RESPONSE	object	filling-in of response command. Different operations have different values. For more information, please refer to the follow-up business introduction

Classification Of Equipment Types

It means classifying equipment. According to different services offered by equipment, equipment can be classified into IPC, MDVR and NVR. The following table content is classified according to different functions of equipment.

After the device is classified, the device of that type is abstracted to obtain the device capability set description. During the device authentication and description phase of communication, the service provider device description is used, and the device requesting the service accesses the device according to the capability set.

Equipment_Type_Table

type mark	equipment name
0	PC(PC or mobile operating equipment)
1	DVR
2	IPC
3	NVR
4	MDVR
5	MIPC
6	DSM
7	P2
8	BWC
9	P3

Link Management

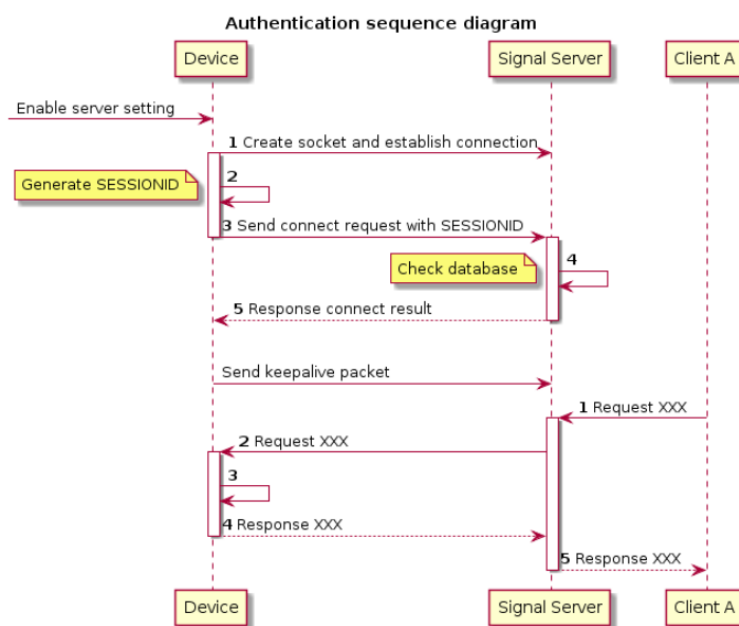
Signaling link

Function Description

Summary

Equipment authentication means establishing communication between the client and equipment, including the client connecting to the equipment. Verify the connection and log in the equipment. The time threshold of this process is 15 seconds. Connection exceeding the time threshold is deemed as illegal connection. The server can reject the connection and disconnect the link. The authentication process generates the unique communication descriptor and subsequent communication requires the descriptor SESSION.

Authentication Process



Description of Authentication Process

1. Firstly, as the client, the device establishes TCP connection with the signaling server and the user can go to the next step after successful connection;
2. The device generates `SESSIONID` which is a unique global descriptor;
3. Send connection command package (includes the device serial number, device type, `SESSIONID` , etc);
4. The server receives connection command and directly returns the error code:
 1. If the central server database includes the device, then the response is successful and save `S0` (the same as `SESSIONID`) as well as global descriptor `SESSIONID` data for subsequent connection. `SESSIONID` is taken as the unique identification of successful auto report. The device serial number in connection command remains the unique identification of the device;
 2. If the central server database doesn't include the device, then the error code needs to be returned and the link be disconnected.

The Process of General Business

1. When the client has business demands about the device, it can send request to the signaling server (custom protocol can be used for business interaction between the client and signaling server);
2. The server forwards request to the device through the protocol;
3. The device sends response message to the signaling server;
4. The signaling server forwards message to the client;

The Description of Re-connection

- If the device fails to receive response from the server within 5s, then old link needs to be disconnected and send new link be sent to the server;
- To reduce re-connection to the server caused by the device disconnection (network anomaly or heartbeat timeout), 3 non-stop re-connections (are called re-connection business. Any one of 3 re-connections is successful, indicating successful re-connection business) are required after the disconnection. If 3 re-connections all fail, the device needs to wait 45s for new re-connection.

Heartbeat

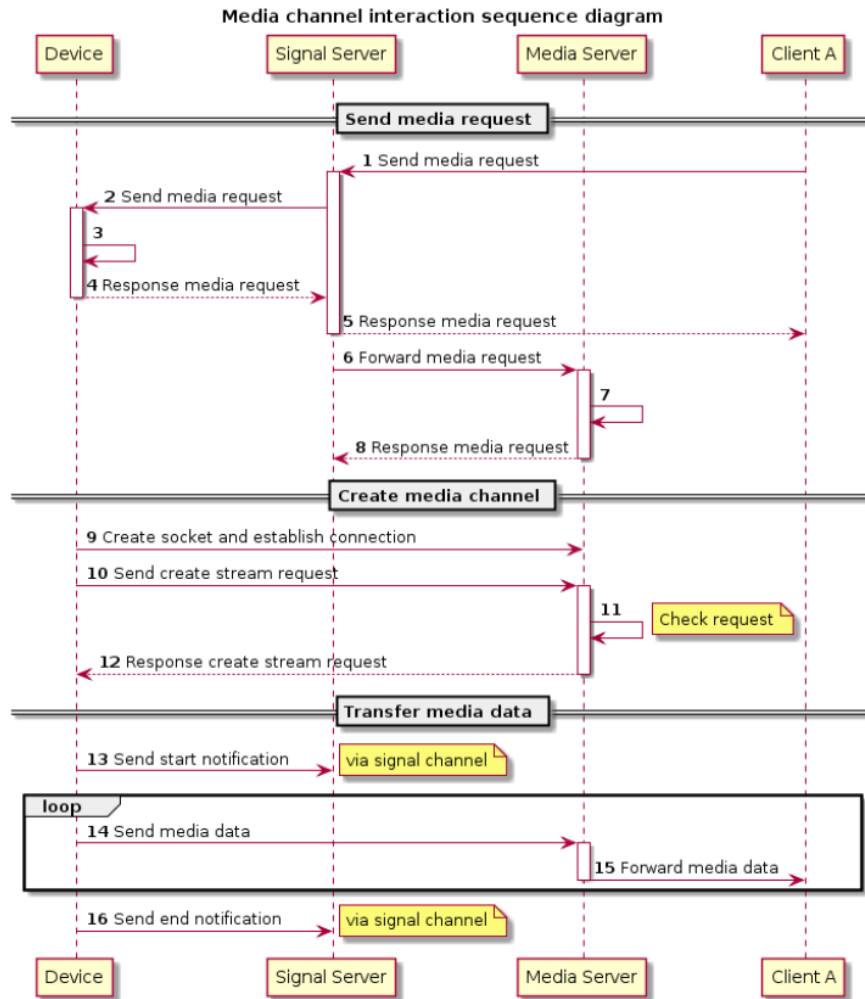
The device sends keep-alive command to the server so as to keep connection. The time interval of heartbeat package is to trigger every 45s. No response after 5 consecutive heartbeats are deemed as disconnected.

Streaming Media Link

Function Description

The transmission content of streaming media includes streaming data such as audio & video photos and files as well as the description of streaming data. The transmission channel is independent of the signaling channel and called as media channel.

Interaction Process



Description of Interaction Process

1. The client sends a command to the signaling server to request the media business from an online device (the device has been online on the signaling server through the signaling link process);
2. The signaling server forwards the media request to the online device;
3. The device checks the request parameter and gives the response to the signaling server. If the device allows the media request, you can go to the next step, otherwise, the device process will be terminated;
4. The signaling server forwards the media request to the client and you can go to the next step if the forwarding is successful;
5. To relieve the pressure on the media server, the signaling can bring about the media server address when requesting the media data. The device will prioritize to use the address in the signaling command to upload information to the media server;
6. The signaling server can forward the successful media request to the media server for processing;
7. After the device successfully connects to the media server, creates streaming media channel command for creating the media channel through . The media channel is officially established after receiving the successful response;
8. The device sends Media signaling session start notification command to the server through the media channel, notifying the media task corresponding to the server to start transmission;
9. The device sends the media data to the media server through the media channel and the media server distributes the media data. Since specific media businesses are different (such as preview, video download and playback), the interaction and data format are different after successfully creating the media channel. For more details, please refer to corresponding chapters;
10. The device sends Media signaling session end notification command to the server through the media channel, notifying the media task corresponding to the server to end transmission.

Connection Command

Function Description

The device initiates the authentication process.

Protocol Introduction

MODULE

CERTIFICATE

OPERATION

CONNECT

PARAMETER

Attribute	Type	Value range	Description
NET	INTEGER		The network used by the device for connection: 0-Wired network, 1-WIFI and 2-3G/4G. You can judge the type to ensure whether to perform the task that consumes data traffic, such as download and preview
DEVNAME	STRING		The device name, which can be displayed to indicate the device type
CARNUM	STRING		License plate number of the device
CID	STRING		Customer Identifier
ICCID	STRING		ICCID of SIM card
AUTONO	STRING		Self-number of the device
AUTOCAR	STRING		Self-number of the vehicle
LINENO	STRING	Less than 32 characters	Line number
UNO	STRING	Less than 32 characters	Driver ID
UNAME	STRING	Less than 32 characters	Driver name
CPN	STRING	Less than 32 characters	Company name
EID	STRING		WIFI essid
DEVCLASS	INTEGER		Device class. Refer to the 'Protocol Encapsulation-Device Type Table

DEVCLASS	INTEGER		Device class. Refer to the 'Protocol Encapsulation-Device Type Table
PRO	STRING		Current protocol version number
CHANNEL	INTEGER		The number of device channel, which indicates the maximum of video channels supported by the current N9M host
CNAME	STRING		Customer name, if the field does not exist, it defaults to standard product
MTYPE	INTEGER		Main device type number
STYPE	INTEGER		Device subtype number. If the field does not exist, it defaults to no subtype.
DEVTYPE	INTEGER		Product type, if the field does not exist, it defaults to regular product 1: Regular product 2: Low-end product 3: 365 customer project product
TSE	INTEGER		Whether the device supports the time shifting function: 0-No and 1-Yes. If the field doesn't exist, it indicates not support
EV	STRING	Less than 100 characters	Indicates the uploading version of evidence supported by the device. "V1.0":Device active upload "V1.1":Platform request "V2.0":Platform request + HTTP upload
FSV	INTEGER		File system version information: 0-4.0 file system and 1-5.0 file system. If the field doesn't exist, the default is the 4.0 file system
PV	INTEGER		Platform version. If the field does not exist, the default value is 1.0 platform 0: N9M 1.0 1: N9M 2.0
SC	INTEGER		Connect to the server subscript at this time, starting from 0
DSNO	STRING		The device serial number. It is used for vehicle management. The value is encrypted chip number. Each N9M host has different chip numbers and the chip number is the unique identifier for the device

RESPONSE

Attribute	Type	Value range	Description
ERRORCODE	INTEGER		refer to the definition of error code
ERRORCAUSE	STRING	less than 100 characters	refer to the definition of error cause
DEVTYPE	INTEGER	1~3	
PRO	STRING		
VCODE	STRING		
MASKCMD	INTEGER		Represent the protocol that the server supports to response when bit value is '1', which is used to realize the transmission of history data on device: Bit0:alarm Bit1:Scanning Gun Bit2:Passenger flow statistics Bit3:Face contrast Bit4:Card swiping Report Bit5:Shutdown Report Bit6:GPS Report

Example

Request

```
{
  "MODULE": "CERTIFICATE",
  "OPERATION": "CONNECT",
  "SESSION": "SDFF9893-LSLIU-KJLJN-JFLJLKJL",
  "PARAMETER": {
    "NET": 2,
    "EV": "V1.1",
    "DEVNAME": "X5AHD",
    "CARNUM": "123456",
    "CPN": "CHEMITO",
    "DSNO": "001A001B001C",
    "UNO": "ST00072",
    "UNAME": "Tony",
    "CHANNEL": 12
  }
}
```

Response

```
{
  "MODULE": "CERTIFICATE",
  "OPERATION": "CONNECT ",
  "SESSION": "SDFF9893-LSLIU-KJLJN-JFLJLKJL",
  "RESPONSE": {
    "MASKCMD": 1,
    "ERRORCODE": 0,
    "ERRORCAUSE": "SUCCESS"
  }
}
```

Keep Alive Command

Function Description

It works as a heartbeat command used for keeping the connection.

Protocol Introduction

MODULE

CERTIFICATE

OPERATION

KEEPALIVE

Example

Request

```
{
  "MODULE": "CERTIFICATE",
  "OPERATION": "KEEPALIVE",
  "SESSION": "SDFF9893-LSLIU-KJLJN-JFLJLKJL"
}
```

Response

```
{
  "MODULE": "CERTIFICATE",
  "OPERATION": "KEEPALIVE",
  "SESSION": "SDFF9893-LSLIU-KJLJN-JFLJLKJL"
}
```

Creates Streaming Media Channel Command

Function Description

Request streaming media service to automatically create streaming media channel. To create the channel, `SESSIONID` must be sent through the channel. The server confirms the channel has been successfully established after receiving `SESSIONID`. The link manages to be added to the session, otherwise, the link is directly disconnected. After successful establishment, the channel can send two-way media streaming. Since then, `STREAMNAME` is the unique identification for each media channel (for example, the same TCP link can be represented by streamname).

Protocol Introduction

MODULE CERTIFICATE

OPERATION CREATESTREAM

PARAMETER

Attribute	Type	Value range	Description
STREAMNAME	STRING		generated by the client. When sending the media business request, the name of media link needs to have this field.
DEVTYPE	INTEGER	1~3	Product Type, field does not exist defaults to Regular Product 1: Regular Product 2: Low-end Product 3: 365 Customer Program Product
DSNO	STRING	1~32 bytes	Device serial number, unique
VISION	STRING	1~32 bytes	Protocol version

RESPONSE

Attribute	Type	Value range	Description
ERRORCODE	INTEGER		refer to the definition of error code
ERRORCAUSE	STRING	less than 100 characters	refer to the definition of error cause

Example

Request

```
{
  "MODULE": "CERTIFICATE",
  "OPERATION": "CREATESTREAM",
  "PARAMETER": {
    "DEVTYPE": "",
    "DSNO": "00B5000053",
    "STREAMNAME": "4-500",
    "VISION": "1.0.4"
  },
  "SESSION": "0000001008206F2F212156B31CF66AA2"
}
```

Response

```
{
  "MODULE": "CERTIFICATE",
  "OPERATION": "CREATESTREAM",
  "SESSION": "SDFF9893-LSLIU-KJLJN-JFLJLKJL",
  "RESPONSE": {
    "ERRORCODE": 0,
    "ERRORCAUSE": "SUCCESS"
  }
}
```

Start Notification of Media Channel Session

Function Description

After establishing the media task (media application command has been dealt with), media session starts. At this time, notify the receiver to prepare for starting media session through the channel.

Protocol Introduction

MODULE

MEDIASTREAMMODEL

OPERATION

MEDIATASKSTART

PARAMETER

Attribute	Type	Value range	Description
CSRC	STRING	0-64 bytes	service provider
PT	INTEGER		for details, please refer to protocol encapsulation
SSRC	INTEGER	0-65535	same as the parameter import application command
STREAMNAME	STRING	1-32 characters	the media channel name
IPANDPORT	STRING	1-32 characters	the media address, the format is <code>IP address: port number</code> , like <code>192.168.1.1:1234</code>

RESPONSE

Attribute	Type	Value range	Description
ERRORCODE	INTEGER		refer to the definition of error code
ERRORCAUSE	STRING	less than 100 characters	refer to the definition of error cause

Example Request

```
{
  "MODULE": "MEDIASTREAMMODEL",
  "OPERATION": "MEDIATASKSTART",
  "PARAMETER": {
    "CSRC": "",
    "IPANDPORT": "192.168.161.79:8002",
    "PT": 2,
    "SSRC": 0,
    "STREAMNAME": "4-500"
  },
  "SESSION": "0000001008206F2F212156B31CF66AA2"
}
```

End Notification of Media Channel Session

Function Description

When the media task is over, notify the receiver to end the media task through the signaling link.

Protocol Introduction

MODULE

MEDIASTREAMMODEL

OPERATION

MEDIATASKSTOP

PARAMETER

Attribute	Type	Value range	Description
ERRORCODE	INTEGER		refer to the definition of error code
ERRORCAUSE	STRING	less than 100 characters	refer to the definition of error cause
CSRC	STRING	0-64 bytes	service provider
IPANDPORT	STRING	1-32 characters	the meida address, the format is IP address: port number , like 192.168.1.1:1234
PT	INTEGER		for details, please refer to protocol encapsulation
SSRC	INTEGER	0~65535	same as the parameter import application command
STREAMNAME	STRING	1-32 characters	the media channel name

Example Request

```
{
  "MODULE": "MEDIASTREAMMODEL",
  "OPERATION": "MEDIATASKSTOP",
  "PARAMETER": {
    "CSRC": "",
    "ERRORCAUSE": "SUCCESS",
    "ERRORCODE": 0,
    "IPANDPORT": "192.168.161.79:8002",
    "PT": 2,
    "SSRC": 0,
    "STREAMNAME": "4-500"
  },
  "SESSION": "0000001008206F2F212156B31CF66AA2"
}
```

Device Status

General Device Status Query

Function Description

The command is to query the current status of the device, including server connection, Bluetooth, 4G, etc.

Command interaction: send the query command to the device through the signaling link and the device responds to it after receiving the command

Protocol Introduction

MODULE

DEVEM

OPERATION

QUERYDEVGENERALSTATUS

RESPONSE

Attribute	Type	Value range	Description
WLM[N]	OBJECT ARRAY		Communication module status information. Currently, the maximum of modules supported by the device is 2
WFM[N]	OBJECT ARRAY		WIFI module status information, which is shown in an array. Array0 indicates internal modules and array1 means external modules
GPSM	OBJECT		Positioning module status information
BTM	OBJECT		Bluetooth module status
CHS[N]	OBJECT ARRAY		Channel status information. The array number indicates the maximum of channels supported by the device.
STOR[N]	OBJECT ARRAY		Storage device status information. The number of storage supported by each device varies and the maximum is 10
VKS	OBJECT		Device key status
IOS[N]	OBJECT ARRAY		IO status information. The array number equals to the maximal supported IO number
CMSC[N]	OBJECT ARRAY		Central server connection status. N is the number of server currently set. The maximum cannot be more than the maximum of connected server
CANBOX	OBJECT		CAN-BOX module status
NETTYPE	INTEGER	0~2	Network connection type: 0-Wired, 1-WIFI, 2-Communication module(2G/3G/4G etc)
SERIAL	INTEGER		It is an unsigned interger and valid in the forward mode. Its MSB is 0 and it is the same as the request

Communication Module Status WLM

Attribute	Type	Value range	Description
EN	INTEGER	0~1	The module enable: 0-on and 1-off
MS	INTEGER	0~1	The module status: 0-existent and 1-non-existent
SIMS	INTEGER	0~3	SIM card status: 0-non-existent, 1-existent, 2-invalid and 3-valid
BHS	INTEGER	0~6	Dialing status: 0-unknown, 1-dialing, 2-successful dial-up, 3-connection exception, 4-the center denies the service due to failed verification, 5-stop dialing and 6-unknown dialing error
IMEI	STRING		IMEI number
IMSI	STRING		IMSI number
NM	INTEGER	0~8	Network standard 0: unknown 1: GPRS, 2: CDMA, 3: EVDO, 4: WCDMA, 5: EDGE, 6: TDSCDMA, 7: LTE-TDD and 8: LTE-FDD
NT	INTEGER	0~4	Current network type: 0-no service, 1-mixed mode, 2-2G, 3-3G and 4-4G
NN	STRING		The name of network type
IP	STRING		It is the IP address of dialing after successful dial-up, otherwise, the value is 0.0.0.0
SVER	STRING	Less than 64 characters	Module software version number, which is shown in a string and directly displayed
RSSI	INTEGER		Signal strength indication. It is a negative number (the unit is db)
MN	INTEGER	0~17	Module model: 0-unknown, 1-U6300, 2-C5300, 3-U8300, 4-U8300W, 5-U8300C, 6-MU709s-2, 7-MU709s-6, 8-ME909s-821, 9-ME909s-120, 10-MS43B, 11-MC7354, 12-EM350, 13-ME3760, 14-LE910, 15-SLM630B, 16-SIM7100C and 17-SIM7100CE
ICCID	STRING		The ICCID of SIM card

WIFI Module Status WFM

Attribute	Type	Value range	Description
EN	INTEGER	0~2	The module enable: (only valid to internal WIFI module information) 0-off, 1-station on and 2-AP on
MD	INTEGER	0~1	WIFI mode: 0-station and 1-AP
MS	INTEGER	0~6	The module status: when MD=0, 0: indicates successful connection, 1: no WIFI module, 2: WIFI module detected, 3: verifying, 4: failed verification, 5: successful verification, 6: acquiring IP address (dhcp). When MD=1, 0: AP successfully enabled, 1: no WIFI module and 2: WIFI module detected
ID	STRING		MD=0 indicates AP ESSID to be verified and MD=1 means ESSID of itself
SL	INTEGER	1~5	MD=0: the signal value1-5. MD=1: invalid
NN	STRING		The name of network type

Positioning Module Status Information GPSM

Attribute	Type	Value range	Description
GS	INTEGER	0~3	Positioning status: 0-valid positioning data, 1-invalid positioning data and 2-no positioning module
GT	INTEGER	0~3	Positioning mode: 0-GPS, 1-Beidou, 2-GLONASS and 3-Galileo
GPN	INTEGER		GPS satellite number
BDN	INTEGER		Beidou satellite number
GLN	INTEGER		GLONASS satellite number
GAN	INTEGER		Galileo satellite number
CTS	INTEGER	0~2	Antenna status: 0-normal, 1-open circuit and 2-short circuit

Bluetooth Device List Information BTM.CONNLIST

Attribute	Type	Value range	Description
STAT	INTEGER	0~2	Connection status: 0-not connected, 1-matched and 2-connected
NAME	STRING		The name of the connected device
ADDRESS	STRING		Address of the connected module

Channel Status CHS

Attribute	Type	Value range	Description
CAMS	INTEGER	0~2	Camera status: 0-normal, 1-not connected and 2-not enabled
RECS	INTEGER	0~3	Recording status: 0-no recording in main stream and sub-stream, 1-no recording in main stream while normal recording in sub-stream, 2-normal recording in main stream while no recording in sub-stream, 3-normal recording in main stream and sub-stream

Storage Device Status STOR

Attribute	Type	Value range	Description
SV	INTEGER	0~1	Validity of the storage: 0-invalid and 1-valid
ST	INTEGER	0~4	Storage type: 0-HDD, 1-SD card, 2-USB, 3-ESARA and 4-array device
SS	INTEGER	0~4	Storage status: 0-no storage, 1-unformatted, 2-full HDD, 3-not recording and 4-recording
SIDX	INTEGER		Serial number of the storage device
SP	INTEGER	0~1	Storage location: 0-internal and 1-external

HDD Lock Status Of Car Key VKS

Attribute	Type	Value range	Description
CKS	INTEGER	0~1	Car key status: 0-off (flameout) and 1-on (ignition)
HDK	INTEGER	0~1	HDD lock status: 0-off (unlocked) and 1-on (locked)

IO Status Information IOS

Attribute	Type	Value range	Description
NAME	STRING		IO name
NSER	STRING	Less than 8 characters	Name abbreviation
SNO	INTEGER		IO number, starting from 1
STATUS	INTEGER		IO status: 0-low level, 1-high level, 2-open circuit and 3-short circuit
USG	INTEGER	0~140	IO purpose: 0-null, 1-emergency alarm, 2-call, 3-neutral gear, 4-air conditioning, 5-turn on the meter, 7-environmental-friendly cover, 8-front door, 9-front door is open, 10-front door is closed, 11-middle door 1, 12-middle door 1 is open, 13-middle door 1 is closed, 14-back door 1, 15-back door 1 is open, 16-back door 1 is closed, 17-low beam headlights, 18-high beam headlights, 19-right turn, 20-left turn, 21-braking, 22-safety commitment, 23-reset, 24-reversing, 25-anti-forgetting alarm, 26-motion detection peripheral alarm, 27-alarm for door opening & closing, 28-safety belt, 29-back cover, 30-IP intercom, 32-stop alarm, 33-boarding and alighting, 34-alarm for covered video, 35-driver alarm, 36-passenger alarm, 37-secondary engine, 39-lift, 40-top cover, 41-electric motor, 42-vacant, 43-PTO, 44-pulse door 4 is open, 45-pulse door 4 is closed, 46-door 4, 47-door 5, 48-door 6, 49-door 7, 50-door 8, 51-hand-pressing alarm button, 52-step-on alarm button, 53-manual intelligent capture, 54-panic alarm, 55-anti-tamper alarm of the host, 56-anti-tamper alarm of the controller, 60-loudspeaker (whistle), 61-smoke alarm, 62-parking brake, 63-driver inspection, 64-protective cover of emergency exit, 65-emergency exit, 68-remote preview (police use and can be custom), 69-GPS report (police use and can be custom), 70-fog lamp, 72-clutch, 73-low oil pressure alarm, 74-alarm for low air pressure I, 75-alarm for low air pressure II, 76-Geo-fence, 77-camera switch (X3-SE252074), 78-driving interference alarm, 82-door 5 is open, 83-door 5 is closed, 84-door 6 is open, 85-door 6 is closed, 86-door 7 is open, 87-door 7 is closed, 88-door 8 is open, 89-door 8 is closed, 90-UPS connects to the vehicle power, 91-left flip, 92-right flip, 93-lift completely, 100-left door of the front row, 101-right door of the front row, 102-service evaluation (dissatisfied), 103-service evaluation (satisfied), 104-service evaluation (very satisfied), 105-headlight signal, 106-driver position, 107-co-pilot position, 108-left position of back row, 109-right position of back row, 110-left door of back row, 111-right door of back row, 112-trunk, 113-the meter is on, 114-engine, 115-nozzle, 116-nozzle, 117-parking, 118-flushing sensor, 119-water volume sensor, 120-life detector, 121-blocking violation, 122-left front door, 123-left middle door, 124-left rear door, 125-ordinary ticket, 126-priority card, 127-children's ticket coin, 128-ordinary ticket coin, 129-card swipe data checkout, 130-summary clear, 131-shift card swipe, 132-shift clear, 133-freeze or blockade, 134-sound and light alarm, 135-passenger flow statistics summary data clear, 136-manual capture of bus lane occupation, 137-radar front obstacle detection, 138-buffer, 139-full load, 140-platform capture

Central Server Connection Status CMSC

Attribute	Type	Value range	Description
IDX	INTEGER		Serial number of the server, starting from 0
EN	INTEGER	0~1	Whether the server is enabled: 0-not enabled and 1-enabled
PTO	INTEGER		For the server protocol type, please refer to parameter documents: 0-N9M protocol, 2-808 protocol and 3-operation and maintenance protocol
STATUS	INTEGER		Server connection status: 0-not connected, 1-connected to the main server, 2-connected to the backup server, -1-the vehicle has been registered, -2-no such vehicle in the database, -3-the terminal has been registered, -4-no such terminal in the database, -5-unencrypted server and -6-failed server verification
ADDR	STRING		The address and port of server registration, such as "192.168.20.110:5556"
NT	INTEGER	0~4	Types of network used by the server: 0-wired network, 1-WIFI, 2-communication module 1, 3-communication module 2 and 4-self-adaptive

CAN-BOX Module Information CANBOX

Attribute	Type	Value range	Description
V	STRING	Less than 32 characters	Version number
CS	INTEGER	0~1	Connection status: 0-not connected and 1-connected

Example

The following is the example of querying I/O and storage together. The status can be queried separately or queried together. SESSION in the example is the session ID sent from the device when connection with the device is established for the first time.

Request

```
{
  "MODULE": "DEVEMM",
  "OPERATION": "QUERYDEVGENERALSTATUS",
  "SESSION": "SDF9893-LSLIU-KJLJN-JFLJLKJL",
  "PARAMETER": {
    "QMASK": 160,
    "SERIAL": 1
  }
}
```

Response

```
{
  "MODULE": "DEVEMM",
  "OPERATION": "QUERYDEVGENERALSTATUS",
  "SESSION": "SDF9893-LSLIU-KJLJN-JFLJLKJL",
  "RESPONSE": {
    "STOR": [
      { "SV": 1, "ST": 3, "SS": 4, "SIDX": 0, "SP": 0 },
      { "SV": 0, "ST": 1, "SS": 2, "SIDX": 1, "SP": 0 }
    ],
    "IOS": [
      { "NAME": "Sensor1", "NSER": "S1", "SNO": 1, "STATUS": 0, "USG": 0 },
      { "NAME": "Sensor2", "NSER": "S2", "SNO": 2, "STATUS": 0, "USG": 0 },
      { "NAME": "Sensor3", "NSER": "S3", "SNO": 3, "STATUS": 0, "USG": 0 }
    ]
  }
}
```

Device Status Query

Function Description

The command is to query the current status of the device.

Protocol Introduction

MODULE

DEVEMM

OPERATION

GETDEVINFOSTATUS

PARAMETER

Attribute	Type	Value range	Description
SERIAL	INTEGER		It is an unsigned interger and valid in the forward mode. Its MSB is 0 and it is the same as the request

RESPONSE

Attribute	Type	Value range	Description
ALARM	INTEGER		Alarm type, refer to the Alarm Status Code Table
BV	INTEGER	100~7200	Device battery voltage, keep 2 valid decimals and enlarge the number by 100 times
G3	ENUM	0~3	3G status. 0: module non-existent. 1: SIM card non-existent. 2: neither module nor SIM card existent. 3: both module and SIM card existent.
G3S	ENUM	0~5	3G signal strength. 0: no signal; 5:strongest signal
G4	ENUM	0~3	4G status. 0: module non-existent. 1: SIM card non-existent. 2: neither module nor SIM card existent. 3: both module and SIM card existent.
G4S	ENUM	0~5	4G signal strength. 0: no signal; 5:strongest signal
W	ENUM	0~1	WIFI status. 0:non-existent; 1:existent
WS	ENUM	0~5	WIFI signal strength. 0: no signal; 5:strongest signal
V	INTEGER	100~7200	Device voltage, keep 2 valid decimals and enlarge the number by 100 times. For example, 100 means 1.00V; 7200 means 72.00V
TD	INTEGER	0~9999, 10000~19999	Device temperature, keep 2 valid decimals and enlarge the number by 100 times. Centigrade below zero is represented by 5 digits. For example, 999 means 9.99°C, 10999 means -9.99°C
TC	INTEGER	0~9999, 10000~19999	Temperature in vehicle, keep 2 valid decimals and enlarge the number by 100 times. Centigrade below zero is represented by 5 digits. For example, 999 means 9.99°C, 10999 means -9.99°C
S	INTEGER	0~99999	Vehicle speed, keep 2 valid decimals and enlarge the number by 100 times. For example, 999 means 9.99 KM/H
SU	ENUM	0~1	Unit of speed. 0:KM/H; 1:MI/H
SW	ENUM	0~1	Car key status: 0-off (flameout) and 1-on (ignition)
RE[N]	ENUM	0~3	Recording status, N represents channel number: 0-no recording in main stream and sub-stream, 1-no recording in main stream while normal recording in sub-stream, 2-normal recording in main stream while no recording in sub-stream, 3-normal recording in main stream and sub-stream

T	STRING	14 bytes	Local time in the format "YYMMDDHHMMSS"
STC	INTEGER	greater than 0	Number of storage
STCINFO[N]	OBJECT ARRAY		Storage status info.
TRAFFIC[N]	OBJECT ARRAY		Device net traffic info.
VS[N]	ENUM		Video loss status. N represents channel number. 0: no video loss. 1: video loss.
H	INTEGER		Humidity, enlarge the number by 10000 times
WD	ENUM	0~16	(Not supported yet)Wind direction. 0-no wind,1-north 2-north by northeast 3-northeast 4- east by northeast 5-east 6-east by southeast 7-southeast 8-south by southeast 9-south 10-south by southwest 11-southwest 12-west by southwest 13-west 14-west by northwest 15-northwest 16-north by northwest
TM	STRING		Total mileage(KM)
HTR	ENUM		Hard disk heating status. 0:normal;1: heating

Storage Status Info STCINFO

Attribute	Type	Value range	Description
T	ENUM	0~2	Storage type. 0:hard disk; 1:u disk; 2: sd card
O	ENUM	0~1	Storage location. 0:internal; 1:external
S	ENUM	0~1	Storage status. 0:normal; 1:fault
TS	INTEGER		Storage total size.
LS	INTEGER		Storage available size.
DS	ENUM		Detailed status. 0: no storage; 1: unformatted; 2: formatting; 3: unmount; 4: storage full; 5: normal; 6: recording; 7: testing; 8: write error; 9: partition error

Device Net Traffic Info TRAFFIC

Attribute	Type	Value range	Description
T	ENUM	0~3	Traffic type. 0:wired; 1: WIFI; 2: 3G; 3: 4G-LTE
I	STRING	less than 32 bytes	NIC IMSI number
TS	INTEGER		Monthly NIC total traffic(KB)
TX	INTEGER		Monthly NIC send traffic(KB)
RX	INTEGER		Monthly NIC receive traffic(KB)

Example Request

```
{
  "MODULE": "DEVEMM",
  "OPERATION": "GETDEVINFOSTATUS",
  "PARAMETER": {
    "SERIAL": 1
  }
}
```

Response

```
{
  "MODULE": "DEVEMM",
  "OPERATION": "GETDEVINFOSTATUS",
  "RESPONSE": {
    "ERRORCAUSE": "SUCCESS",
    "ERRORCODE": 0,
    "S": {
      "ALARM": 0,
      "BV": 0,
      "G3": 1,
      "G3S": 0,
      "G4": 2,
      "G4S": 0,
      "H": 0,
      "HTR": 0,
      "RE": [2, 0, 0, 0, 0, 0, 0],
      "S": 0,
      "SINFO": [
        {"DS": 6, "LS": 1862405193728, "O": 0, "S": 0, "T": 0, "TS": 1999844147200},
        {"DS": 5, "LS": 58376323072, "O": 0, "S": 0, "T": 2, "TS": 63484985344},
        {"DS": 0, "LS": 0, "O": 0, "S": 1, "T": 1, "TS": 0},
        {"DS": 0, "LS": 0, "O": 0, "S": 1, "T": 1, "TS": 0},
        {"DS": 0, "LS": 0, "O": 0, "S": 1, "T": 1, "TS": 0}
      ],
      "STC": 5,
      "SU": 0,
      "SW": 1,
      "T": "20220627095758",
      "TC": 3600,
      "TD": 4200,
      "TM": "67437.658000",
      "V": 1880,
      "VS": [0, 1, 1, 1, 1, 1, 1, 1],
      "W": 0,
      "WS": 0
    },
    "SERIAL": 1
  }
}
```

IO Car Key Status

Acquire IO and Car Key Status

Function Description

The command is to search the device's I/O status and no longer supported after 2020. Please refer to the General Device Status Query chapter

Protocol Introduction

MODULE

DEVEMM

OPERATION

GETUPDATEIOSTATUS

PARAMETER

Attribute	Type	Value range	Description
SERIAL	INTEGER		It is an unsigned integer, valid in the forward mode and its MSB is 0

RESPONSE

Attribute	Type	Value range	Description
SERIAL	INTEGER		It is an unsigned integer, valid in the forward mode and its MSB is 0
IO	OBJECT ARRAY	The array index is I/O serial number	sensor input status
ACC	OBJECT		ACC status
FN	OBJECT		Pulse information
PB	INTEGER	0~1	Panic button status: 0-not occurred and 1-occurred

Sensor Input Status of IO IO

Attribute	Type	Value range	Description
NAME	STRING		Name
NSER	STRING		Name abbreviation
S	INTEGER	0~3	IO status: 0-low level, 1-high level, 2-open circuit and 3-short circuit
U	INTEGER		For the IO purpose, please refer to the description of IO status information (IOS)

ACC Status Parameters ACC

Attribute	Type	Value range	Description
A	INTEGER	0~1	Ignition. 0-flameout and 1-ignition

Pulse Information FN

Attribute	Type	Value range	Description
N	INTEGER		The pulse number, which should be an integer

Example

Request

```
{
  "MODULE": "DEVEMM",
  "OPERATION": "GETUPDATEIOSTATUS",
  "SESSION": "SDFF9893-LSLIU-KJLJN-JFLJLKJL",
  "PARAMETER": {"SERIAL": 1}
}
```

Response

```
{
  "MODULE": "DEVEMM",
  "OPERATION": "GETUPDATEIOSTATUS",
  "SESSION": "SDFF9893-LSLIU-KJLJN-JFLJLKJL",
  "RESPONSE": {
```

```
"SERIAL":1,
"IO":[{"NAME" : "SENSOR IN 1","USER":"IO1","S":0, "U" : 1},{ "NAME" : "SENSOR IN 2","USER":"IO2","S":1, "U" : 2}],
"ACC":{"A" : 1},
"FN":{"N":1000},
"FB":0
}
}
```

Actively Upload IO and Car Key Status

Function Description

When the device's IO status changes, the device actively pushes a message to the signaling server.

Protocol Introduction

MODULE
DEVEMM

OPERATION
UPDATEIOSTATUS

PARAMETER

Attribute	Type	Value range	Description
IO	OBJECT ARRAY	The array index is I/O serial number	sensor input status
ACC	OBJECT		ACC status
FN	OBJECT		Pulse information
FB	INTEGER	0~1	Panic button status: 0-not occurred and 1-occurred

Example

```
{
"MODULE":"DEVEMM",
"OPERATION":"UPDATEIOSTATUS",
"SESSION":"SDFF9893-LSLIU-KJLJN-JFLJLKJL",
"PARAMETER":{
"IO":[{"NAME" : "SENSOR IN 1","USER":"IO1","S":0, "U" : 1},{ "NAME" : "SENSOR IN 2","USER":"IO2","S":1, "U" : 2}],
"ACC":{"A" : 1},
"FN":{"N":1000},
"FB":0,
"P":{"V":0,"J":2320.123456, "W":1030.123456,"S":6000,"C":3000,"T":"20191015085901","H":300}
}
}
```

Special Report

Function Description

When the PT value in the protocol header is 22, it indicates a special command on the signaling channel between the device and server so as to facilitate the server to handle some special command faster. PT=22 offers a special path. SSRC in the protocol header determines the meaning of different commands and its meaning description is decided by the following table. The PT value must be unique and its maximum is 65535

SSRC Table

SSRC	Description
0	Heartbeat is used to initiate the heartbeat interaction protocol between the device and the signaling server. It replaces the original heartbeat instruction KEEPALIVE (3.2. 2). The direct connection mode still uses KEEPALIVE. It only contains protocol header data, no protocol body data, and the heartbeat strategy does not change.
1	Heartbeat is used for the heartbeat interactive response protocol between the device and the signaling server. The original heartbeat command KEEPALIVE (3.2. 2) response is abolished, and the direct connection mode still uses KEEPALIVE. It only contains protocol header data, no protocol body data, and the heartbeat strategy does not change.
2	GPS data upload

GPS Data Structure

```
typedef struct Net_GPS
{
    char viled; //
    //0 indicates subsequent GPS-related data is reliable and valid;
    //1 indicates data might not be reliable. It usually happens when there is an insufficient number of tracked satellites;
    //2 indicates the device has no GPS module.
    char uexpand; //It indicates whether the location information has been expanded: 0-not expanded. Package it according to the current field. (for more details, please refer to GPSEX field in 3.4.5.79 Temporary Start & Stop Function Protocol)
    char ureal; //The data type: 0-real-time data and 1-supplementally-uploaded data
    char reserver[1];
    int ulongitude; //Longitude. MSB 0 indicates east longitude and 1 means west longitude. Keep 6 valid decimals and enlarge the number by 1 million times. For example, 150000890 indicates 150.00089°E. Use endianness and the range is 179999999 to 1179999999, indicating 179.999999°E to 179.999999°W.
    int ulatitude; //Latitude. MSB 0 indicates north latitude and 1 means south latitude. Save 6 valid decimals and enlarge the number by 1 million times. For example, 1150089000 indicates 150.089°S. Use endianness and the range is 179999999 to 1179999999, indicating 179.999999°N to 179.999999°S.
    int uspeed; //Ground speed, which is shown in a positive integer. Keep 2 valid decimals and enlarge the number by 100 times. For example, 99999 indicates the current speed is 999.99km/h (the unit is km/h). Use endianness and the value range is 0-99999, indicating the speed ranges from 0km/h to 999.99km/h.
    int udirect; //Ground driving direction, which is an integer. Keep 2 valid decimals and enlarge the value by 100 times. For example, 18090 indicates 180.90°. Take due north as reference and use clockwise rotation. Use endianness. The range is 0-35999, indicating 0° to 359.99°
    int uhigh; //Altitude (relative to the sea level). A negative number indicates below the sea level and a positive number mean above the sea level. The range is -32768 to 32767 and endianness is used.
    char utime[16]; //The current time with time zone, which is shown in a string. For example, year/month/day/hour/minute/second: 20120928121212, indicating 12:12:12, 28th , September, 2012
    char exdatabody[0]; //Expanded data. It is valid when "uexpand" is 1. The extended data follows the GPS data.The 3837 client is an FTSDData_t structure.
}
```

Customer 3922 Customized

```
typedef struct GPSExtendCustom3922
{
    CustomAExGps_t customAExGps;
    unsigned int ssr; //Temporarily useless, take the SSRC in the protocol header
    unsigned int cmdno; //Unique identifier, returned in response
}ExGpsCustom3922_t;
```

Bus India 3962 Customer Base Station Information Structure

```
typedef struct DTC_BaseStation
{
    short length; /* Base station information length, four-byte alignment, network transmission, total length of body*/
    short realen; /* The actual length of the base station information content, the length of the valid data in the body*/
    char body[0]; // Base station information content, Eg: "404, 11, 040a, 9139, 56, 913a, 040a, 43, b385, 040a, 38, d17a, 040a, 37, 32ab, 040a"
}DTCBaseStation;

typedef struct stuNetGpsExSensorBoardData
{
    char header[3]; //Frame Header
    char analogData[31]; //RI Input Temperature Value
    char pulseData[20]; //Connecting the oil level sensor
    char digitalInput[3]; //Digital Input
    char checkBit[2]; //Check digit
    char reserve1[65]; //RX1 input for connecting to RFID board, CAN, bus and other peripherals
    char reserve2[13]; //RX2 input for connecting to RFID board, CAN, bus and other peripherals
    rm_uint8_t end; //Frame end
}NetGpsExSensorBoardData;
```

Indian customer 61260 customized tire pressure data structure

```
typedef struct stuTPMSItemData
{
    mm_uint16_t TirePressure; //Tire pressure: Unit: 0.1 bar, 0x3FF is an invalid value, and the tire pressure may not be displayed
    mm_int16_t TireTemperature; //Temperature: Unit: Degree Celsius, 0xFF is invalid value, tire temperature may not be displayed.
    mm_uint8_t TireAlarm; //0: No alarm, 1: Alarm, bit0: Tire pressure too high, bit1: Tire pressure too low
    bit2: Tire temperature is too high, bit3: Slow leak alarm
    bit4: quick leakage alarm, bit5: low battery alarm
    mm_uint8_t TireValidStatus; //Tyre valid status, 0: invalid, 1: valid
    mm_uint8_t reserved[2];
}NetTPMSItemData_t;
typedef struct stuTPMSData
{
    mm_uint8_t ItemCount; //Number of tires, maximum 14 tires
    mm_uint8_t ucReserved[3];
    NetTPMSItemData_t ItemData[14]; //Tire pressure data
}NetGpsExTPMSData_t;
//GPS supplementary transmission response: add serial number upload data structure, which is also used as platform response data structure.
typedef struct stuGpsExNormalAck
{
    unsigned int ssrc;//1. When the device reports, the SSRC is the same as the SSRC in the protocol header and is 2; 2. When the platform responds,
    the SSRC is fixed to 12.
    unsigned int cmdno;//Same as reported
}NetGpsExNormalAck_t;
```

3807 Customer uploads power-on reason for models with power box

```
typedef struct stuNetGpsExPowerOnReason
{
    xuint8_t powerboxReason; // The reason for the power box to start. 0-initialization loading, 1-Gsensor start, 2-OBDD ACC ON start, 3-hard line
    ACC start
    xuint8_t reserved[3]; // Reserve
}NetGpsExPowerOnReason_t;
//General peripheral data transparent transmission structure
typedef struct
{
    xuint16_t length; //Body field length
    xuint8_t reserved[2];
    xbyte_t body[0]; //Transparent data
}GpsExCommonDispatcherData_t;
```

OBD data reporting

OBD ID	ID Description	Value Description
1	OBD speed	2 significant decimal places reserved, value expanded by 100 times, unit KM/H
2	Engine speed	2 significant decimal places reserved, value expanded by 100 times, unit RPM
3	Total mileage	2 significant decimal places reserved, value expanded by 100 times, unit KM
4	Average fuel consumption	2 significant decimal places reserved, value expanded by 100 times, unit KM/L
5	Gear	0:D, 1:R, 2:N, 3:P
6	Vehicle left and right turning status	0:No turn, 1:Left turn, 2:Right turn
7	Brake (foot brake)	0: No brake; 1: Brake
8	Steering angle	2 significant decimal places reserved, value expanded by 100 times, unit degree
9	Battery voltage	2 significant decimal places reserved, value expanded by 100 times, unit V
10	Driver seat belt	0: connected, 1: not connected
11	Throttle position	Percentages are expressed with 2 decimal places and the value is multiplied by 100. For example, 6000 means 60%
12	High beam	0: Off, 1: On
13	Engine operating time	unit: second
14	Engine coolant temperature	2 significant decimal places reserved, value expanded by 100 times, unit °C
15	Oil pressure	2 significant decimal places reserved, value expanded by 100 times, unit KPa
16	Engine intake air flow	2 significant decimal places reserved, value expanded by 100 times, unit g/s
17	Engine load	2 significant decimal places reserved, value expanded by 100 times, unit %
18	Clutch status	0: Clutch not pressed; 1: Clutch pressed
19	Remaining oil	2 significant decimal places reserved, value expanded by 100 times, unit %
20	Ignition status	0: OFF, 1: ON
21	Instantaneous fuel consumption	2 significant decimal places reserved, value expanded by 100 times, unit L/H
22	Total fuel consumption	2 significant decimal places reserved, value expanded by 100 times, unit L
other	To be expanded	

```

typedef struct stuObdItem
{
    xuint16_t obdId; //Type Id
    xuint8_t reserved[2];
    xint32_t value; //Data value, signed integer value
}ObdItem_t
typedef struct stuNetGpsExObdData
{
    xuint8_t version; //Version information, fixed to 0
    xuint8_t authStatus; //Authorization status, 0: unauthorized, 1: authorized
    xuint8_t analyseStatus; //Parsing status, 0: not connected, 1: part of the parsing is normal, 2: successful, 3: parsing failed
    xuint8_t reserved[5]; //Reserve
    xuint16_t dataCount; //The number of data items, indicating how many ObdItem_t there are
    xuint8_t reserved1[2]; //Reserve
    xbyte data[0]; //OBD data, total length is dataCount*sizeof(ObdItem_t)
}NetGpsExObdData_t;
typedef struct
{
    unsigned short type; //Extended data type, 0: Vehicle information 1: Cement truck sensor status 2: GPS advanced,
    3: Customized by customer 3922, 4: Customized by customer 133036 - carrying multiple speed information;
    5. Thailand 52074 Project equipment basic information and fuel quantity information report 6. Public transport India 3962 Customer base station
    information,
    7. Customer 133036 customized - transparent transmission of SensorBoard docking data 8. Indian customer 61260 customized tire pressure data
    9. Standard GPS serial number data reporting, 10. 205628 customer transparent data; 11. 3807 Customers upload the reasons for starting the
    machine with a power box
    12. Transparent transmission of general peripheral data 13. Alcohol tester data from Japanese customers 14. OBD data
    char reserverd[2];
    int length; // Specifies the actual length of the following data body
    union // This only shows the supported data body definition set. Do not use this union definition to calculate the length directly.
    {
        NetGpsExVehicleInfo vehicleInfo;
        NetGpsExTruckSensor sensorInfo;
    }
}

```

```

NetGpsExGPSAdvanced gpsAdvanced;
ExGpsCustom3922_t exGpsCustom3922;
NetGpsExSpeedsInfo speedsInfo;
NetGpsExBangkoDeviceInfo deviceInfo; //52074 basic information, fuel information
DTCTBaseStation baseStationInfo; //Base station information
NetGpsExSensorBoardData sensorBoardData;
NetGpsExTPMSData_t TPMSData;
NetGpsExNormalAck_t normalAckInfo; //Extend ack to add serial number to upload
NetGpsExTransmission_t transparentInfo; //Transparent data is reported along with GPS
NetGpsExPowerOnReason_t powerOnReason; //Reasons for starting the machine with a power box
GpsExCommonDispatcherData_t dispatcherData; //General transparent data
NetGpsExAlcoholData _t alcoholData; //Alcohol tester data
NetGpsExObdData_t obdData; //OBD data
}NetGpsExContent;
}GpsExtendFourth; //GPS Extended Protocol 4

```

Device Management

Function Description

This command is used to control the state of the device.

Protocol Introduction

MODULE

DEVEMM

OPERATION

SETCONTROLDEVCMMD

PARAMETER

Attribute	Type	Value range	Description
CMDTYPE	ENUM	0~2	Command type. 0:restart;1:sleep poweroff;2:poweroff.
SERIAL	INTEGER		Command serial number.
CONTINUE	INTEGER		Shutdown time in seconds. If the value is 0 it means permanent shutdown. If it is greater than 0 it means the time interval between the next power on and this shutdown (maximum support 65535*60). This parameter is valid when CMDTYPE is 1.

RESPONSE

Attribute	Type	Value range	Description
ERRORCODE	INTEGER		0: success 11: The protocol currently does not support this function: 1. The CMDTYPE field is not 0, 1, or 2;
ERRORCAUSE	STRING	less than 100 characters	refer to the definition of error cause
CMDTYPE	ENUM	0~2	Command type. Same as the field in request.
SERIAL	INTEGER		Command serial number. Same as the field in request.

Example

Request

```

{
  "MODULE": "DEVEMM",
  "OPERATION": "SETCONTROLDEVCMMD",
  "PARAMETER": {
    "CMDTYPE": 0,
    "SERIAL": 1
  }
}

```


}

Response

```
{  
  "MODULE": "DEVEMM",  
  "OPERATION": "SETCONTROLDEVCMD",  
  "RESPONSE": {  
    "CMDTYPE": 0,  
    "ERRORCAUSE": "SUCCESS",  
    "ERRORCODE": 0,  
    "SERIAL": 1  
  },  
  "SESSION": "0002878E9599936D0B8A46786448D2A9"  
}
```

Alarm Report

General Alarm Report Format

Protocol Introduction

MODULE

EVEM

OPERATION

SENDALARMINFO

PARAMETER

Attribute	Type	Value range	Description
ALARMTYPE	INTEGER		Alarm type
CMDTYPE	ENUM	0~3	Command type: 0-remove alarm, 1-start alarm, 2-warning, 3-alarm duration
ALARMUID	INTEGER	1~0x7FFFFFFF	indicate whether the alarm is first given or repeatedly sent. If the field of two pieces of alarm information of the same type is the same, then the alarm is repeatedly sent, otherwise, it is a new alarm. The count starts from 1 after the number reaches 0x7FFFFFFF
EVTUUIID	STRING	the maximum of 64 bytes	alarm event ID
RUN	INTEGER	0x0~0xFFFFFFFF	the number of device reboot. The platform cannot accurately search every piece of alarm since the alarm ID needs to be recalculated after the device reboot. Therefore, add this type to form a unique alarm identification with ALARMUID. The identification can be compatible with the previous alarm and is 0 by default on the platform.
ALARMAS	INTEGER	0~2	alarm level: 0-important events, 1-general alarm and 2-emergency alarm
ALARMCOUNT	INTEGER	1~65535	the number of how many times one alarm is sent. For the first time, the number is 1 by default and the number will be accumulated if the alarm is repeatedly sent
TRIGGERTYPE	ENUM	0~1	trigger types: 0-manual and 1-automatic
CONTINUETIME	INTEGER		alarm duration
CURRENTTIME	INTEGER		the time of alarm occurred (UTC with the time zone)
L	INTEGER	0~15	the number of language types, referring to the configuration file: 0-Simplified Chinese, 1-English, 2-Korean, 3-Italian, 4-German, 5-Thai, 6-Turkish, 7-Portuguese, 8-Spanish, 9-Romanian, 10-Greek, 11-French, 12-Russian, 13-Dutch, 14-Hebrew and 15-Traditional Chinese
P	OBJECT		GPS location information
TO	INTEGER		indicate the platform to report alarm to. 0 or the default is N9M platform
REAL	INTEGER	0~1	Indicates whether the upload is real-time or not. 0-real-time upload, 1-supplemental upload; field not present defaults to real-time upload
CMDNO	UNSIGNED INTEGER		command serial number
SECNO	INTEGER		the alarm serial number at the same time
ALARMMODE	INTEGER	0~1	0 or none-single alarm and 1-several alarms combined
PUSH	ENUM	0~1	1-32 (hexadecimal shown in bit. bit0-bit31 indicates channel1-channel 32. The channel alarm requires push when the corresponding bit is 1, otherwise, push is not required. Field not present means no push required)

extend...The above part is the common fields of alarm reporting, and different alarm types hold specific alarm fields, which will be described below

GPS Location Information

Attribute	Type	Value range	Description
V	INTEGER		Positioning status: 0 indicates the following GPS-related data is reliable and valid; 1 indicates the data might not be reliable. It usually happens when there are insufficient satellites tracked and 2 indicates the device has no GPS
J	STRING		Longitude, which is shown in a string of floating-point number. Keep 6 decimal places and the format is dddd.mmmmmm (0 before and after the decimal point shall be kept). The dddd range is -179~179 and mmmmmm range is 0~999999. A positive number indicates east longitude and a negative number indicates west longitude.
W	STRING		Latitude, which is shown in a string of floating-point number. Keep 6 decimal places and the format is dddd.mmmmmm (0 before and after the decimal point shall be kept). The ddd range is -89~89 and mmmmmm range is 0~999999. A positive number indicates north latitude and a negative number indicates south latitude.
S	INTEGER	0~99999	Speed. The unit is 0.01 km/h
C	INTEGER	0~35999	Driving direction. The range is 0°-359.99°. Take due north as 0° and the degree increases clockwise
T	STRING	1~14	Local time with time zone. For example, year/month/day/hour/minute/second: 20120928121212, indicating 12:12:12, 28th September 2012

RESPONSE

Attribute	Type	Value range	Description
ERRORCODE	INTEGER		refer to the definition of error code
ERRORCAUSE	STRING	less than 100 characters	refer to the definition of error cause
ALARMTYPE	INTEGER		alarm type, same as the ALARMTYPE value in PARAMETER
CMDTYPE	ENUM	0~3	command type, same as the CMDTYPE value in PARAMETER
ALARMUID	INTEGER		alarm ID, same as the ALARMUID value in PARAMETER
RUN	INTEGER		the number of device reboot, same as the RUN value in PARAMETER
CMDNO	UNSIGNED INTEGER		command serial number, same as the CMDNO value in PARAMETER

Example

Requeset

as follows...

Response

```
{
  "MODULE": "EVE",
  "OPERATION": "SENDALARMINFO",
  "RESPONSE": {
    "ALARMTYPE": 1,
    "ALARMUID": 1,
    "CMDNO": 61669378,
    "CMDTYPE": 1,
    "ERRORCODE": 0,
    "RUN": 941
  },
  "SESSION": "8b0247ed-db1d-4dbc-80eb-b5b461b26f29"
}
```

Video Loss Alarm

Protocol Introduction

PARAMETER

Attribute	Type	Value range	Description
ALARMTYPE	ENUM	0	video loss alarm
CHANNEL	INTEGER		Channel number, shown in BIT. BIT0-BIT31 means channel 1-channel 32, valid when the bit is 1.
CHANNELMASK	INTEGER		Alarm channel number, shown in BIT. BIT0-BIT31 means channel 1-channel 32, valid when the bit is 1.
LCH[N]	INTEGER		N represents a valid alarm channel number, starting from 0. Each LCH element represents the set of channel numbers linked when the Nth channel alarms, and the set is represented by BIT. For example: when the channel alarm is linked to 1~5 channel recording, the value of this element is 31 (11111 in binary)
ALARMNAME	STRING	"Video Loss"	the alarm name
SER	STRING	"VL"	the abbreviation of alarm name

Example

```
{
  "MODULE": "EVEM",
  "OPERATION": "SENDALARMINFO",
  "PARAMETER": {
    "ALARMAS": 1,
    "ALARMCOUNT": 1,
    "ALARMNAME": "Video Loss",
    "ALARMTYPE": 0,
    "ALARMUID": 4,
    "CHANNEL": 1,
    "CHANNELMASK": 1,
    "CMDNO": 1558970369,
    "CMDTYPE": 1,
    "CURRENTTIME": 1656514970,
    "EVTUUUID": "66ba5f89-07e7-4972-86f0-fe69459c18d3",
    "L": 0,
    "LCH": [
      31
    ],
    "P": {
      "C": 0,
      "J": "0.000000",
      "S": 0,
      "T": "2000000000000000",
      "V": 2,
      "W": "0.000000"
    },
    "REAL": 0,
    "RUN": 23788,
    "SECNO": 0,
    "SER": "VL",
    "SNO": 0,
    "TRIGGERTYPE": 1
  },
}
```

```
"SESSION": "0001524614FC3EC652C4782F064A9DDD"
}
```

Covered Video Alarm

Protocol Introduction

PARAMETER

Attribute	Type	Value range	Description
ALARMTYPE	ENUM	1	covered video alarm
CHANNEL	INTEGER		Channel number, shown in BIT. BIT0-BIT31 means channel 1-channel 32, valid when the bit is 1.
CHANNELMASK	INTEGER		Alarm channel number, shown in BIT. BIT0-BIT31 means channel 1-channel 32, valid when the bit is 1.
LCH[N]	INTEGER		N represents a valid alarm channel number, starting from 0. Each LCH element represents the set of channel numbers linked when the Nth channel alarms, and the set is represented by BIT. For example: when the channel alarm is linked to 1~5 channel recording, the value of this element is 31 (11111 in binary)
ALARMNAME	STRING	"Video Shield"	the alarm name
SER	STRING	VS	the abbreviation of alarm name

Example

```
{
"MODULE": "EVEM",
"OPERATION": "SENDALARMINFO",
"PARAMETER": {
"ALARMAS": 1,
"ALARMCOUNT": 1,
"ALARMNAME": "Video Shield",
"ALARMTYPE": 1,
"ALARMUID": 5,
"CHANNEL": 1,
"CHANNELMASK": 1,
"CMDNO": 1558970371,
"CMDTYPE": 1,
"CURRENTTIME": 1656518386,
"EVTUUUID": "ec2e971d-db2c-4e49-9b8d-ce9c49382891",
"L": 0,
"LCH": [
7
],
"P": {
"C": 0,
"J": "0.000000",
"S": 0,
"T": "200000000000000",
"V": 2,
"W": "0.000000"
},
"REAL": 0,
"RUN": 23788,
"SECNO": 0,
"SER": "VS",
```

```
"SNO": 0,
"TRIGGER": 1
},
"SESSION": "0001524614FC3EC652C4782F064A9DDD"
}
```

Motion Detection Alarm

Protocol Introduction

PARAMETER

Attribute	Type	Value range	Description
ALARMTYPE	ENUM	2	motion detection alarm
CHANNEL	INTEGER		Channel number, shown in BIT. BIT0-BIT31 means channel 1-channel 32, valid when the bit is 1.
CHANNELMASK	INTEGER		Alarm channel number, shown in BIT. BIT0-BIT31 means channel 1-channel 32, valid when the bit is 1.
LCH[N]	INTEGER		N represents a valid alarm channel number, starting from 0. Each LCH element represents the set of channel numbers linked when the Nth channel alarms, and the set is represented by BIT. For example: when the channel alarm is linked to 1~5 channel recording, the value of this element is 31 (11111 in binary)
ALARMNAME	STRING	"Motion Detection"	the alarm name
SER	STRING	"MD"	the abbreviation of alarm name

Example

```
{
"MODULE": "EVEM",
"OPERATION": "SENDALARMINFO",
"PARAMETER": {
"ALARMAS": 1,
"ALARMCOUNT": 1,
"ALARMNAME": "Video Shield",
"ALARMTYPE": 1,
"ALARMUID": 5,
"CHANNEL": 1,
"CHANNELMASK": 1,
"CMDNO": 1558970371,
"CMDTYPE": 1,
"CURRENTTIME": 1656518386,
"EVTUUUID": "ec2e971d-db2c-4e49-9b8d-ce9c49382891",
"L": 0,
"LCH": [
7
],
"P": {
"C": 0,
"J": "0.000000",
"S": 0,
"T": "2000000000000000",
"V": 2,
"W": "0.000000"
},
"REAL": 0,
"RUN": 23788,
```

```
"SECNO": 0,
"SER": "VS",
"SNO": 0,
"TRIGGERTYPE": 1
},
"SESSION": "0001524614FC3EC652C4782F064A9DDD"
}
```

Motion Detection Alarm

Protocol Introduction

PARAMETER

Attribute	Type	Value range	Description
ALARMTYPE	ENUM	2	motion detection alarm
CHANNEL	INTEGER		Channel number, shown in BIT. BIT0-BIT31 means channel 1-channel 32, valid when the bit is 1.
CHANNELMASK	INTEGER		Alarm channel number, shown in BIT. BIT0-BIT31 means channel 1-channel 32, valid when the bit is 1.
LCH[N]	INTEGER		N represents a valid alarm channel number, starting from 0. Each LCH element represents the set of channel numbers linked when the Nth channel alarms, and the set is represented by BIT. For example: when the channel alarm is linked to 1~5 channel recording, the value of this element is 31 (11111 in binary)
ALARMNAME	STRING	"Motion Detection"	the alarm name
SER	STRING	"MD"	the abbreviation of alarm name

Example

```
{
"MODULE": "EVEM",
"OPERATION": "SENDALARMINFO",
"PARAMETER": {
"ALARMAS": 1,
"ALARMCOUNT": 1,
"ALARMNAME": "Motion Detect",
"ALARMTYPE": 2,
"ALARMUID": 8,
"CHANNEL": 1,
"CHANNELMASK": 1,
"CMDNO": 1558970377,
"CMDTYPE": 1,
"CURRENTTIME": 1656519468,
"EVTUUUID": "4710d2e8-f011-47df-aebd-8ab8e2246382",
"L": 0,
"LCH": [
31
],
"P": {
"C": 0,
"J": "0.000000",
"S": 0,
"T": "2000000000000000",
"V": 2,
"W": "0.000000"
},
}
```

```
"REAL": 0,
"RUN": 23788,
"SECNO": 0,
"SER": "MD",
"SNO": 0,
"TRIGGERTYPE": 1
},
"SESSION": "0001524614FC3EC652C4782F064A9DDD"
}
```

Memory Exception Alarm

Protocol Introduction

PARAMETER

Attribute	Type	Value range	Description
ALARMTYPE	INTEGER	3	memory exception alarm
SER	STRING	ST	the abbreviation of alarm name
STORAGETYPE	ENUM	0~2	Storage device type: 0-HDD, 1-USB flash disk and 2-SD card
STORAGEINDEX	INTEGER	1~32	Logical number of the storage
ERRORCODE	ENUM	0~7	Abnormal reason. 0-full disk. 1-unformatted. 2-read/write error. 3-no record. 4. no storage. 5-unmounted. 6-partition error 7-over capacity

Example

```
{
"MODULE": "EVEM",
"OPERATION": "SENDALARMINFO",
"PARAMETER": {
"ALARMAS": 1,
"ALARMCOUNT": 1,
"ALARMTYPE": 3,
"ALARMUID": 6,
"CMDNO": 1558970373,
"CMDTYPE": 1,
"CURRENTTIME": 1656518700,
"ERRORCODE": 4,
"EVTUUUID": "26543c12-bb1f-4412-a237-34091ad1ee8a",
"L": 0,
"P": {
"C": 0,
"J": "0.000000",
"S": 0,
"T": "2000000000000000",
"V": 2,
"W": "0.000000"
},
"REAL": 0,
"RUN": 23788,
"SECNO": 0,
"STORAGEINDEX": 2,
"STORAGETYPE": 2,
"TRIGGERTYPE": 1
},
"SESSION": "0001524614FC3EC652C4782F064A9DDD"
}
```



```
}
```

IO Alarm

Protocol Introduction

PARAMETER

Attribute	Type	Value range	Description
ALARMTYPE	INTEGER	4	IO alarm
SNO	CHAR	1~255	Customized io alarm is numbered by the device to differentiate io alarms. The number mustn't be repeated
ALARMNAME	STRING	less than 32 characters	Alarm name
SER	STRING	IO	abbreviation of alarm name
LCH[N]	INTEGER		N represents a valid alarm channel number, starting from 0. Each LCH element represents the set of channel numbers linked when the Nth channel alarms, and the set is represented by BIT. For example: when the channel alarm is linked to 1~5 channel recording, the value of this element is 31 (11111 in binary)
USE	INTEGER		purpose of IO alarm: 0-null; 1-emergency alarm; 2-calling; 3-neutral gear; 4-air conditioning; 5-run the meter; 6-exceed the height limit; 7-environmental-protection cover; 8-front door; 9-front door is open; 10-front door is closed; 11-middle door 1; 12-middle door 1 is open; 13-middle door 1 is closed; 14-back door 1; 15-back door 1 is open; 16-back door 1 is closed; 17-low-beam light; 18-high-beam light; 19-right turn; 20-left turn; 21-braking; 22-safety commitment and 23-anti-forgetting alarm
L	STRING		The line number, which is shown in string. The field is valid when USE is 23
PC	INTEGER		The number of passengers. The field is valid when USE is 23. The value is the sum of DN and MN
DN	STRING ARRAY		The driver card number. Each element indicates one driver card number. The field is valid when USE is 23
MN	STRING ARRAY		The administrator card number. Each element indicates one administrator card number. The field is valid when USE is 23

Example

```
{
  "MODULE": "EVEM",
  "OPERATION": "SENDALARMINFO",
  "PARAMETER": {
    "ALARMAS": 1,
    "ALARMCOUNT": 1,
    "ALARMNAME": "Sensor1",
    "ALARMTYPE": 4,
    "ALARMUID": 9,
    "CMDNO": 1558970379,
    "CMDTYPE": 1,
    "CURRENTTIME": 1656520062,
    "EVTUUUID": "eebfdd97-920f-45f8-abe2-eceb8369fe69",
    "L": 0,
    "LCH": [
      31
    ],
    "P": {
      "C": 0,
      "J": "0.000000",
      "S": 0,
      "T": "2000000000000000",
      "V": 2,
      "W": "0.000000"
    },
    "REAL": 0,
    "RUN": 23788,
    "SECNO": 0,
    "SER": "S1",
    "SNO": 1,
    "TRIGGERTYPE": 1,
    "USE": 3
  },
  "SESSION": "0001524614FC3EC652C4782F064A9DDD"
}
```

Emergency Alarm

Protocol Introduction

PARAMETER

Attribute	Type	Value range	Description
ALARMTYPE	INTEGER	7	Emergency alarm
ALARMNAME	STRING	"Urgent"	Alarm name
SER	STRING	"UR"	Abbreviation of alarm name
LCH[N]	INTEGER		The channel number of linked video, which is shown in bit. It is valid when bit is 1

Example

```
{
  "MODULE": "EVEM",
  "OPERATION": "SENDALARMINFO",
```

```
"PARAMETER": {
  "ALARMAS": 1,
  "ALARMCOUNT": 1,
  "ALARMNAME": "Urgent",
  "ALARMTYPE": 7,
  "ALARMUID": 23,
  "CMDNO": 1559232567,
  "CMDTYPE": 1,
  "CURRENTTIME": 1656588809,
  "EVTUUUID": "8e0cb876-a9a4-4cfd-8804-cb872bc5cc55",
  "L": 0,
  "LCH": 0,
  "P": {
    "C": 0,
    "J": "0.000000",
    "S": 4100,
    "T": "2000000000000000",
    "V": 2,
    "W": "0.000000"
  },
  "REAL": 0,
  "RUN": 23792,
  "SECNO": 0,
  "SER": "Urgent",
  "TRIGGERTYPE": 1
},
"SESSION": "0000000F2077D9D1798F2F865841415F"
}
```

Speeding Alarm Protocol Introduction

PARAMETER

Attribute	Type	Value range	Description
ALARMTYPE	ENUM	8	Speeding alarm
ATYPE	INTEGER		Alarm type. 0: standard low-speed alarm; 1: standard high-speed alarm; 2: speeding warning; 3: instant high-speed alarm; 4: rapid deceleration alarm; 5: speed recovery alarm; 6: parking; 7: threshold alarm and 8: start alarm (speed changes from 0 to a number other than 0)
CSP	INTEGER	0~99999	Current speed value. The unit is 0.01 km/h
MINSP	INTEGER	0~99999	Minimum speed. The unit is 0.01 km/h
MAXSP	INTEGER	0~99999	Maximum speed. The unit is 0.01 km/h
MINS	INTEGER	0~99999	Actual minimum speed reached in the low-speed alarm. The unit is 0.01 km/h
MAXS	INTEGER	0~99999	Actual maximum speed reached in the high-speed alarm. The unit is 0.01 km/h
ALARMNAME	STRING	less than 32 characters	Alarm name
SER	STRING	PRESPD/SPD	Abbreviation of alarm name
AT	CHAR		Location type. 0: no specific location; 1: circular area; 2: rectangular area; 3: polygon area and 4: road section;
I	INTEGER		If P field exists, it indicates the corresponding area or road section ID, otherwise, it is invalid
LD	INTEGER		Speeding alarm level: 0: no level; 1: P1 level 1; 2: P1 level 2; 3: P1 level 3; 4: P1 level 4 and 5: P1 level 5;

Example

```
{
  "MODULE": "EVEM",
  "OPERATION": "SENDALARMINFO",
  "PARAMETER": {
    "ALARMAS": 1,
    "ALARMCOUNT": 1,
    "ALARMNAME": "Over Speed",
    "ALARMTYPE": 8,
    "ALARMUID": 11,
    "AT": 0,
    "ATYPE": 1,
    "CMDNO": 1558970383,
    "CMDTYPE": 1,
    "CONTINUE TIME": 0,
    "CSP": 7700,
    "CURRENTTIME": 1656522240,
    "EVTUUUID": "7705e06e-5a50-431a-b825-7292ab51eb4a",
    "I": 0,
    "L": 0,
    "LD": 5,
    "MAXS": 77,
    "MAXSP": 5000,
    "MINS": 0,
    "MINSP": 0,
    "P": {
      "C": 0,
      "J": "0.000000",
      "S": 7700,
      "T": "2000000000000000",
      "V": 2,
      "W": "0.000000"
    },
    "REAL": 0,
    "RUN": 23788,
    "SECNO": 0,
    "SER": "SPD",
    "TRIGGER TYPE": 1
  },
  "SESSION": "0001524614FC3EC652C4782F064A9DDDD"
}
```

Low Voltage Alarm

Protocol Introduction

PARAMETER

Attribute	Type	Value range	Description
ALARMTYPE	INTEGER	9	Voltage alarm
SER	STRING	LV	Abbreviation of alarm name
V	INTEGER	100~7200	Current voltage. Enlarge the voltage by 100 times and keep 2 decimal places
S	ENUM	0~1	voltage alarm type. 0: low voltage alarm; 1: high voltage alarm

Example

```
{
  "MODULE": "EVE",
  "OPERATION": "SENDALARMINFO",
  "PARAMETER": {
    "ALARMAS": 1,
    "ALARMCOUNT": 1,
    "ALARMTYPE": 9,
    "ALARMUID": 5,
    "CMDNO": 1558970379,
    "CMDTYPE": 1,
    "CURRENTTIME": 1656519700,
    "EVTUUUID": "d4cc9f84-c99d-430f-9ec6-4a37ef136678",
    "L": 0,
    "P": {
      "C": 0,
      "J": "0.000000",
      "S": 0,
      "T": "2000000000000000",
      "V": 2,
      "W": "0.000000"
    },
    "REAL": 0,
    "RUN": 23788,
    "S": 0,
    "SECNO": 0,
    "TRIGGERTYPE": 1,
    "V": 6200
  },
  "SESSION": "0001524614FC3EC652C4782F064A9DDD"
}
```

Geo Fence Alarm

Protocol Introduction

PARAMETER

Attribute	Type	Value range	Description
ALARMTYPE	INTEGER	17	Alarm for entering & exiting Geo-fence
SER	STRING	AIO	Abbreviation of alarm name
E	INTEGER		Geo-fence event type, which is shown in bit. When the value corresponding to bit is 1, it indicates the corresponding event exists. bit0: enter the area; bit1: exit the area; bit2: enter the line; bit3: exit the line; bit4: lane departure; bit5: too long/insufficient driving hour;
AC	INTEGER ARRAY		Action of entering and exiting Geo-fence. Each element of the array indicates an action associated with Geo-fence event. The action is shown in bit and bit indicates: bit0 1: report to the platform; 0: don't report to the platform; bit1 1: door opening is forbidden; 0: door opening is allowed; bit2 1: communication module is off 0: communication module is on; bit3 1: collect detailed GNSS positioning data; 0: don't collect detailed GNSS positioning data;
AT	INTEGER		Geo-fence type: 1-circular, 2-rectangular, 3-polygon and 4-line;
ID	INTEGER		Geo-fence ID
AN	STRING	less than 32 characters	Geo-fence name
GEO	INTEGER		Video status: 0-off and 1-on. When the field is null, it indicates the Geo-fence doesn't control the video
TO	INTEGER		Source number of Geo-fence, starting from 1

Example

```
{
  "MODULE": "EVEM",
  "OPERATION": "SENDALARMINFO",
  "PARAMETER": {
    "ALARMAS": 1,
    "ALARMCOUNT": 1,
    "ALARMTYPE": 17,
    "ALARMUID": 5,
    "CMDTYPE": 1,
    "CURRENTTIME": 1570805579,
    "E": 1,
    "AC": [1],
    "AT": 1,
    "ID": 1,
    "AN": "123",
    "GEO": "0",
    "EVTUUUID": "d4cc9f84-c99d-430f-9ec6-4a37ef136678",
    "L": 0,
    "P": {
      "C": 0,
      "J": "113.998321",
      "S": 0,
      "T": "20191011065257",
      "V": 0,
      "W": "22.596535"
    },
    "RUN": 933,
    "STORAGEINDEX": 0,
    "TRIGGERTYPE": 1,
  },
  "SESSION": null
}
```

ACC Alarm Protocol Introduction PARAMETER

Attribute	Type	Value range	Description
ALARMTYPE	INTEGER	18	ACC alarm
ALARMNAME	STRING	"Acceleration"	Alarm name
SER	STRING	"ACC"	Abbreviation of alarm name
LCH	INTEGER		The linked channel number, which is shown in bit. It is valid when bit is 1
D	INTEGER		Direction: 0-X direction; 1-Y direction; 2-Z direction; 3-collision; 4-rollover; 5-bump; 6-rapid acceleration; 7-rapid deceleration; 8-G4; 9-G5; 10-sharp left turn and 11-sharp right turn
X	INTEGER		Enlarge it by 1000 times
Y	INTEGER		Enlarge it by 1000 times
Z	INTEGER		Enlarge it by 1000 times
V	INTEGER		Valid G4 and G5 alarm

Example

```
{
  "MODULE": "EVEM",
  "OPERATION": "SENDALARMINFO",
  "PARAMETER": {
    "ALARMAS": 1,
    "ALARMCOUNT": 1,
    "ALARMNAME": "Acceleration",
    "ALARMTYPE": 18,
    "ALARMUID": 1,
    "CMDNO": 1559232515,
    "CMDTYPE": 1,
    "CURRENTTIME": 1656525037,
    "D": 0,
    "EVTUUUID": "a6f010f7-48cb-49bb-bb6f-384462747a7c",
    "L": 0,
    "LCH": 0,
    "P": {
      "C": 0,
      "J": "0.000000",
      "S": 1800,
      "T": "2000000000000000",
      "V": 2,
      "W": "0.000000"
    },
    "REAL": 0,
    "RUN": 23792,
    "SECNO": 0,
    "SER": "ACC",
    "TRIGGERTYPE": 1
  },
  "SESSION": "0000000F2077D9D1798F2F865841415F"
}
```

DSM AI Alarm

Protocol Introduction

PARAMETER

Attribute	Type	Value range	Description
ALARMTYPE	INTEGER	56	DSM(AI) alarm
SER	STRING	DSM	Abbreviation of alarm name
ST	INTEGER		Alarm sub-types
SP	INTEGER		Location of sensor
LCH	INTEGER		The channel number of linked video, which is shown in bit. It is valid when bit is 1.
LEV	INTEGER		Alarm level: 1-level-1 alarm and 2-level-2 alarm;
LDWTYPE	INTEGER		The field is valid when ST is 5 (lane departure alarm). Different numbers indicate different directions: 0-left and 1-right;
KW	STRING		The field of the keyword of uncivilized words is valid when ST is 30
LSP	STRING		This field is valid when ST is 7. Indicates the identified speed limit value

DSM Alarm Sub-types

Value	Description	Value	Description
0	Driver fatigue	1	No driver
2	Driver's phone calling	3	Driver smoking
4	Driver distraction	5	lane departure
6	Forward collision	7	Speeding warning
8	Lisence plate recognition	9	Too close following distance
10	Yawning	11	Pedestrian collision
12	Aisle overcrowding	13	Pick up passengers outside the stop
14	Illegal lane change	15	Phone playing
16	Seat belt unfastened	17	Driver verification failed
18	Blind spot detection (right)	19	Virtual bumper alarm (urban forward collision alarm)
20	Two hands off the steering wheel	21	one hand off the steering wheel
22	Infrared-blocking glasses	23	blind spot detection (left)
24	Passenger interference alarm	25	intersection speed limit alarm
26	intersection detection alarm	27	Pedestrian detection (crosswalk)
28	Alarm for no courtesy to pedestrians (crosswalk)	29	Crosswalk detection
30	Uncivilized words of the driver		

Example

```
{
  "MODULE": "EVEM",
  "OPERATION": "SENDALARMINFO",
  "PARAMETER": {
    "ALARMAS": 1,
    "ALARMCOUNT": 1,
    "ALARMTYPE": 56,
    "ALARMUID": 10,
    "CMDNO": 1559232542,
    "CMDTYPE": 1,
    "CURRENTTIME": 1656584816,
    "EVTUUUID": "8bc1516b-02c7-400c-b0f2-0d18e6af8161",
    "L": 0,
    "LEV": 1,
    "P": {
      "C": 0,
      "J": "0.000000",
      "S": 4100,
      "T": "2000000000000000",
    }
  }
}
```



```
"V": 2,
"W": "0.000000"
},
"REAL": 0,
"RUN": 23792,
"SECNO": 0,
"SP": 0,
"ST": 1,
"TRIGGERTYPE": 1
},
"SESSION": "0000000F2077D9D1798F2F865841415F"
}
```

Temperature Threshold Alarm

Protocol Introduction

PARAMETER

Attribute	Type	Value range	Description
ALARMTYPE	INTEGER	61	Temperature threshold alarm
ALARMNAME	STRING	1-128	Alarm name
SER	STRING	1-32	Alarm name abbreviations TETH: Temperature alarm PTETH: Temperature warning TEWD: Temperature rise warning
LCH	INTEGER		The channel number of linked video, which is shown in bit. It is valid when bit is 1.
SNO	INTEGER		The number, which starts from 0. 0-3 indicate the number of temperature sensor
CT	INTEGER		Max temperature threshold. Keep two decimal places and enlarge the number by 100 times
SCT	INTEGER		Min temperature threshold. Keep two decimal places and enlarge the number by 100 times
PCT	INTEGER		emperature alarm offset. Keep two decimal places and enlarge the number by 100 times
CWD	INTEGER		Temperature change threshold only available when warming alarm happend. Keep two decimal places and enlarge the number by 100 times
CWT	INTEGER		Warming duration with second unit.

Example

```
{
"MODULE": "EVEM",
"OPERATION": "SENDALARMINFO",
"PARAMETER": {
"ALARMAS": 1,
"ALARMCOUNT": 1,
"ALARMTYPE": 61,
"ALARMUID": 5,
"CMDTYPE": 1,
"CURRENTTIME": 1570805579,
"CT": 3000,
"EVTUUUID": "d4cc9f84-c99d-430f-9ec6-4a37ef136678",
```

```

"L":0,
"LCH": 0,
"P":{
"C":0,
"J": "113.998321",
"S":0,
"T": "20191011065257",
"V":0,
"W": "22.596535"
},
"PCT": 0,
"RUN":933,
"SCT": 4000,
"SNO": 0,
"TRIGGERTYPE":1,
},
"SESSION": "0000000F2077D9D1798F2F865841415F"
}

```

Humidity Alarm

Protocol Introduction

PARAMETER

Attribute	Type	Value range	Description
ALARMTYPE	INTEGER	116	Humidity threshold alarm
ALARMNAME	STRING	1-128	Alarm name
SER	STRING	HDY	Abbreviation of alarm name
SNO	INTEGER		The number, which starts from 0. 0-3 indicate the number of humidity sensor
LCH	INTEGER		The channel number of linked video, which is shown in bit. It is valid when bit is 1.
LHDY	INTEGER		Max humidity threshold. Keep two decimal places and enlarge the number by 100 times
SHDY	INTEGER		Min humidity threshold. Keep two decimal places and enlarge the number by 100 times
PHDY	INTEGER		Humidity alarm offset. Keep two decimal places and enlarge the number by 100 times
CHDY	INTEGER		Current humidity. Keep two decimal places and enlarge the number by 100 times

Example

```

{
"MODULE": "EVEM",
"OPERATION": "SENDALARMINFO",
"PARAMETER": {
"ALARMAS": 1,
"ALARMCOUNT": 1,
"ALARMTYPE": 61,
"ALARMUID": 5,
"CHDY": 3000,
"CMDTYPE": 1,

```

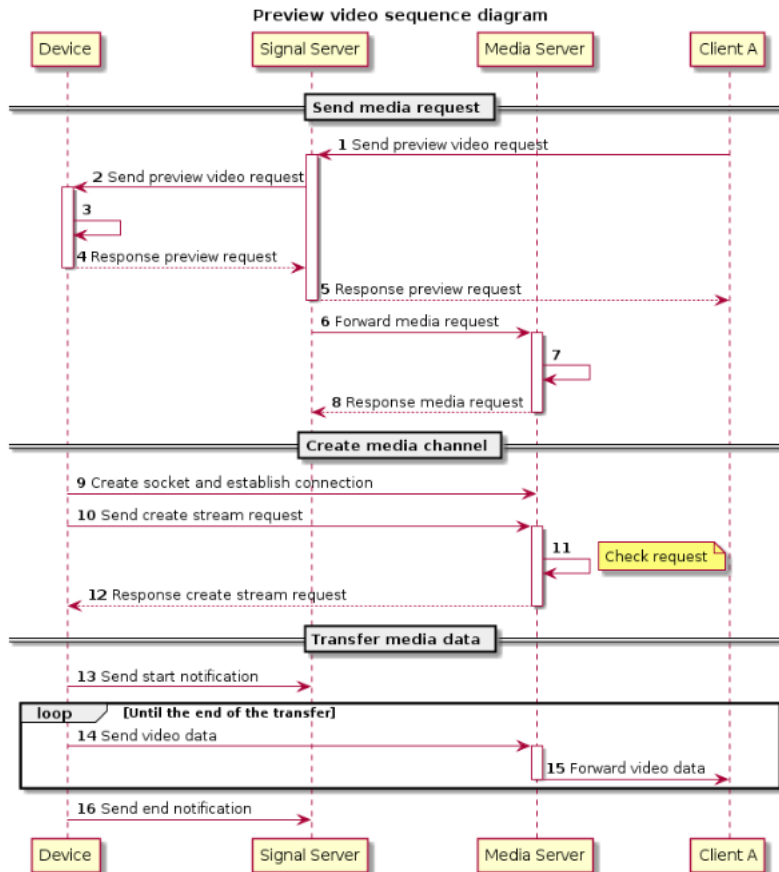
```
"CURRENTTIME":1570805579,
"EVTUUUID":"d4cc9f84-c99d-430f-9ec6-4a37ef136678",
"L":0,
"LHDY": 8000,
"LCH": 0,
"P":{
"C":0,
"J":"113.998321",
"S":0,
"T":"20191011065257",
"V":0,
"W":"22.596535"
},
"PHDY": 0,
"RUN":933,
"SHDY": 7000,
"SNO": 0,
"TRIGGERTYPE":1,
},
"SESSION":"0000000F2077D9D1798F2F865841415F"
}
```

Preview

Real Time Video Preview

Function Description

When a user submits a request for video, if the user already has the same task, there is no need to request it again; if it is a new media task, the media task should be assigned to an appropriate thread for execution.



Steps Explanation

1. The client sends a live video request and the command is forwarded to the device via the signaling server.
2. The device verifies request parameters after receiving live video request command. Preview is allowed if the response is successful and you can go to the next step. Otherwise, the response fails. live video request command
3. The device establishes the media channel. For the procedure, please refer to link management - streaming media link chapter;
4. After the media channel is established, the device sends Media signaling session start notification command via the signaling channel, notifying the media task corresponding to the client to start transmission;
5. The media server receives real-time video streaming on the media channel and forwards (or sends) it to the corresponding client;
6. If data transmission is over or wrong, the device will send Media channel session end notification command via the signaling channel, notifying the media task corresponding to the server to end transmission;

Control Instructions

During the preview, the client can send real-time video control and request for frame I command to control the preview function.

Supported control is shown below:

1. Switch video streaming. The client can request the device to switch video streaming types transmitted by preview. For more information, please refer to Real-time video control - example ;
2. Only upload frame I. The client can request the device to only transmit frame I and discard frame P and frame A. For more information, please refer to Real-time video control - example ;
3. Start & stop audio. The client can request the device to start & stop the audio transmission of any channel. For more information, please refer to Real-time video control - example ;

4. Request for frame I. The client can request the device to immediately generate one frame I. For more information, please refer to Request for frame I - example ;

Real Time Video Request

Function Description

Request for real-time video preview of the device.

Protocol Introduction

MODULE

MEDIASTREAMMODEL

OPERATION

REQUESTALIVEVIDEO

PARAMETER

Attribute	Type	Value range	Description
SSRC	INTEGER	0-255	The field is invalid when the platform requests the video. Fill in channel number(0 indicates channel 1) into SSRC when the device responds
STREAMNAME	STRING	1-32 characters	the media channel name
STREAMTYPE	INTEGER	0-2	Requested video stream types: 0-sub-stream, 1-main stream and 2-mobile phone stream
CHANNEL	INTEGER		Shown in bit. Bit0-bit 31 indicate channel 1 to channel 32. The corresponding channel is valid when bit is 1, otherwise, it is invalid.
AUDIOVALID	INTEGER		It is shown in bit. Bit0-Bit31 indicate Channel 1- Channel 32. The audio is valid when bit is 1. It corresponds to the channel number corresponding to CHANNEL and the corresponding channel number must be valid. Not support now.
IPANDPORT	STRING	1-32 characters	Media address. If there is no such field, then use the ip:port set by the service provider, such as 58.60.231.218:5550. This field is requirement.
FRAMECOUNT	INTEGER	0-30	Network transmission frame rate. The device uploads information to the platform according to the actual frame rate by default. Not support now.
FRAMEMODE	INTEGER	ENUM	The mode used for sending information. The field is empty or 0 by default when non-existent, indicating normal play according to frame rate of the current coding. 0-normal play, 1-discard all frames P and frames A. Not support now.
SERIAL	UNSIGNED INTEGER		Command serial number. It is valid in the forward mode

RESPONSE

Attribute	Type	Value range	Description
ERRORCODE	INTEGER		0: success 25: Task already exists: 1. The current preview task is being executed 26: Resources are not sufficient or the task is full: 1. Shared cache application failed 28: Not encoded: 1. The obtained video channel is not encoded 85: No remote device: 1. No remote device
ERRORCAUSE	STRING	less than 100 characters	refer to the definition of error cause
SSRC	INTEGER	0-255	The field is invalid when the platform requests the video. Fill in CHANNEL into SSRC when the device responds
STREAMNAME	STRING	1-32 characters	the media channel name, the same as the request
STREAMTYPE	INTEGER	0-2	Requested video stream types: 0-sub-stream, 1-main stream and 2-mobile phone stream
SERIAL	UNSIGNED INTEGER		Command serial number. It is valid in the forward mode

Example

Explain

Request to transmit sub-stream real-time audio & video of channel 1 from the device.

Request

```
{
  "MODULE": "MEDIASTREAMMODEL",
  "OPERATION": "REQUESTALIVEVIDEO",
  "PARAMETER": {
    "CHANNEL": 1,
    "IPANDPORT": "192.168.161.79:8002",
    "STREAMNAME": "channel1-sub",
    "STREAMTYPE": 2
  },
  "SESSION": "000000996A3A5E2666A2E8A36B00E397"
}
```

Response

```
{
  "MODULE": "MEDIASTREAMMODEL",
  "OPERATION": "REQUESTALIVEVIDEO",
  "RESPONSE": {
    "ERRORCAUSE": "SUCCESS",
    "ERRORCODE": 0,
    "SSRC": 0,
    "STREAMNAME": "channel1-sub",
    "STREAMTYPE": 2
  },
  "SESSION": "000000996A3A5E2666A2E8A36B00E397"
}
```

Real Time Video Control

Function Description

Control real-time video preview of the device.

Protocol Introduction

MODULE

MEDIASTREAMMODEL

OPERATION

CONTROLSTREAM

PARAMETER

Attribute	Type	Value range	Description
PT	INTEGER		The PT value should be consistent with the Media Channel Session Start Notification command response
SSRC	INTEGER	0-255	the channel number, be consistent with the SSRC returned by the real-time video request
STREAMNAME	STRING	1-32 characters	the media channel name
CMD	ENUM		Medial task control operation: 0-stop, 1-resume, 2-pause, 3-switch video stream, 4-audio management, 5-set frame rate for video stream, 6-change the sending mode. We support 0, 3 and 6 only currently.
STREAMTYPE	INTEGER	0-2	Valid when CMD is 3. Types of video stream to be switched: 0-sub-stream, 1-main stream and 2-mobile phone stream. When CMD is 0, this field must the same as request.
AUDIOVALID	INTEGER		It is shown in bit and valid when CMD is 4. Bit0-Bit31 indicate Channel 1- Channel 32. It is valid when BIT is 1. It corresponds to the channel number corresponding to CHANNEL and the corresponding channel number must be valid
FRAMEMODE	INTEGER		Valid when CMD s 6. The mode is used for sending information. The field is empty or 0 by default when non-existent, indicating normal play according to frame rate of the current coding. 0-normal play, 1-discard all frames P and frames A.

RESPONSE

Attribute	Type	Value range	Description
ERRORCODE	INTEGER		0: success 28: Not encoded: 1. CMD == 3, and the switched channel is not encoded 40: No such task: 1. No task corresponding to streamName is found; 2. Failed to find the buffer of the channel corresponding to SSRC (CMD == 3 or 6)
ERRORCAUSE	STRING	less than 100 characters	refer to the definition of error cause
SSRC	INTEGER	255	the channel number
STREAMNAME	STRING	1-32 characters	the media channel name

Example

Switch Video Stream

Initial request is for sub-stream preview and switched to main stream preview

Request

```
{
  "MODULE": "MEDIASTREAMMODEL",
  "OPERATION": "CONTROLSTREAM",
  "PARAMETER": {
    "CMD": 3,
    "PT": 2,
    "SSRC": 0,
    "STREAMNAME": "channel1-sub",
    "STREAMTYPE": 1
  },
  "SESSION": "000000996A3A5E2666A2E8A36B00E397"
}
```

Response

```
{
  "MODULE": "MEDIASTREAMMODEL",
  "OPERATION": "CONTROLSTREAM",
  "RESPONSE": {
    "ERRORCAUSE": "SUCCESS",
    "ERRORCODE": 0,
    "SSRC": 0,
    "STREAMNAME": "channel1-sub"
  },
  "SESSION": "000000996A3A5E2666A2E8A36B00E397"
}
```

Stop Video Streaming

Initial request is for normal play and now stop playing.

Request

```
{
  "MODULE": "MEDIASTREAMMODEL",
  "OPERATION": "CONTROLSTREAM",
  "PARAMETER": {
    "CMD": 0,
    "PT": 2,
    "SSRC": 0,
    "STREAMNAME": "channel1-sub",
    "STREAMTYPE": 2
  }
}
```



```
},  
"SESSION": "000000996A3A5E2666A2E8A36B00E397"  
}
```

Response

```
{  
  "MODULE": "MEDIASTREAMMODEL",  
  "OPERATION": "CONTROLSTREAM",  
  "RESPONSE": {  
    "ERRORCAUSE": "SUCCESS",  
    "ERRORCODE": 0,  
    "SSRC": 0,  
    "STREAMNAME": "channel1-sub"  
  },  
  "SESSION": "000000996A3A5E2666A2E8A36B00E397"  
}
```

Request For I Frame

Function Description

Request the device to send frame I.

Protocol Introduction

MODULE

AVSM

OPERATION

GETIFRAME

PARAMETER

Attribute	Type	Value range	Description
STREAMTYPE	INTEGER	0-2	Valid when CMD is 3. Types of video stream to be switched: 0-sub-stream, 1-main stream and 2-mobile phone stream
CHANNEL	INTEGER		Shown in bit. Bit0-bit 31 indicate channel 1 to channel 32. The corresponding channel is valid when bit is 1, otherwise, it is invalid.

RESPONSE

Attribute	Type	Value range	Description
ERRORCODE	INTEGER		0: success 6: Execution failed: 1. Request for I frame data failed
ERRORCAUSE	STRING	less than 100 characters	refer to the definition of error cause

Example

Explain

Request main stream of channel 1 to send frame I

Request

```
{
"MODULE": "AVSM",
"OPERATION": "GETIFRAME",
"PARAMETER": {
"STREAMTYPE": 1,
"CHANNEL": 1
},
"SESSION": "575f4fbe-9ca6-4c76-9414-0156adf75aa0"
}
```

Response

```
{
"MODULE": "AVSM",
"OPERATION": "GETIFRAME",
"RESPONSE": {
"ERRORCAUSE": "SUCCESS",
"ERRORCODE": 0,
},
"SESSION": "575f4fbe-9ca6-4c76-9414-0156adf75aa0"
}
```

Intercom

Intercom Function Description

The user sends intercom request command. If the user has the same task, then it is needless for reapplication. If it is a new media task, then it is necessary to assign the media task to proper thread for execution

Procedure Description

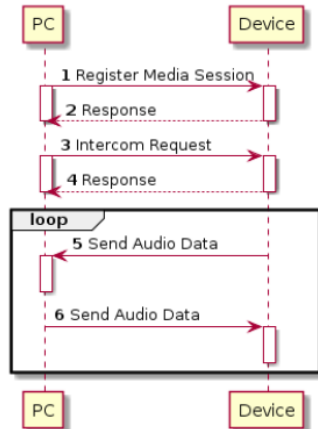
1. The client sends intercom request command and the command is forwarded to the device through the signalling server;
2. The device checks request parameters after receiving intercom request command. If intercom is allowed, then the response is successful and you can go to the next step. Otherwise, the response fails;
3. For the device to establish the media channel, you can refer to Link management - Streaming media link chapter;
4. After establishing the media channel, the device can send media signalling session start notification command on the signalling channel to notify the media task corresponding to the server to start transmission;
5. The media server receives the audio data on the media channel and forwards it to the corresponding client or device;
6. If data transmission is over or wrong, the device will send media signaling session end notification command through the signaling channel, notifying the media task corresponding to the sever to end transmission.

Control Description

During the intercom, the client can send intercom control command to start/stop the function.

Flow Chart

Intercom Interaction Sequence Diagram (Passive Connect)



Intercom Request

Function Description

Intercom request

Protocol Introduction

MODULE

MEDIASTREAMMODEL

OPERATION

REQUESTTALK

PARAMETER

Attribute	Type	Value range	Description
CSRC	STRING	0~64 Bytes	Server identifier
SSRC	INTEGER	0~65535	Command identifier. You need to fill in the identifier for application. The device will enter the identifier in the SSRC field after the successful request
STREAMNAME	STRING	1~32 Bytes	The name of media channel
MODE	INTEGER	0	Intercom/broadcast/monitoring mode. 0: intercom
CHANNEL	INTEGER		It is shown in bit. Bit0-bit31 indicate Channel 1-Channel 32. It is valid when bit is 1. The mode is valid when AUDIOSOURCE=1
SOUNDMODE	INTEGER	0~1	Sound track type and client encoding parameters. 0: mono and 1: stereo
CHANNELTOTAL	INTEGER	1~32	The number of sound track and client encoding parameters
SAMPLINGRATE	INTEGER		Sampling rate and client encoding parameters, such as 8000,11025, 22050,44100
SAMPLINGFIGURE	INTEGER		Sampling bit width and client encoding parameters. 0: 8 bits, 1: 16 bits and 2: 32 bits
AUDIOFORMAT	INTEGER		Audio encoding format and client encoding parameters
AUDIOSOURCE	INTEGER		Encoding parameters of the device audio source specified by the client: 1-camera pick-up and 2-microphone
AUDIOFRAMELEN	INTEGER		Frame lenght of audio and client encoding parameters
IPANDPORT	STRING		Media IP. If the field doesn't exist, then use ip:port set on the device, like 58.60.231.218:5550
SERIAL	INTEGER		serial number

RESPONSE

Attribute	Type	Value range	Description
ERRORCODE	INTEGER		0: success 25: Task exists: 1. The current intercom task is being executed 26: Insufficient resources or task is full: 1. Shared cache application failed
ERRORCAUSE	STRING	less than 100 characters	refer to the definition of error cause
STREAMNAME	STRING	1~32 Bytes	The name of media channel
CHANNEL	INTEGER		It is shown in bit. Bit0-bit31 indicate Channel 1-Channel 32. It is valid when bit is 1. The mode is valid when AUDIOSOURCE=1
SOUNDMODE	INTEGER	0~1	Sound track type and device encoding parameters. 0: mono and 1: stereo
CHANNELTOTAL	INTEGER	1~32	The number of sound track and device encoding parameters
SAMPLINGRATE	INTEGER		Sampling rate and device encoding parameters, such as 8000,11025, 22050,44100
SAMPLINGFIGURE	INTEGER		Sampling bit width and device encoding parameters. 0: 8 bits, 1: 16 bits and 2: 32 bits
AUDIOFORMAT	INTEGER		Audio encoding format and client encoding parameters
AUDIOSOURCE	INTEGER		Device encoding parameters: 1-camera pick-up and 2-microphone
AUDIOFRAMELEN	INTEGER		Frame length of audio and device encoding parameters
SERIAL	INTEGER		The same as the request

Audio_Encoding_Format

Attribute	Value
PCMU	0
1016	1
G721	2
GSM	3
G723	4
DVI4_8K	5
DVI4_16K	6
LPC	7
PCMA	8
G722	9
S16BE_STEREO	10
S16BE_MONO	11
QCELP	12
CN	13
MPEGAUDIO	14
G728	15
DVI4_3	16
DVI4_4	17
G729	18
G711A	19
G711U	20
G726	21
G729A	22
LPCM	23
CelB	25
JPEG	26
CUSM	27
NV	28
PICW	29
CPV	30
H261	31
MPEGVIDEO	32
MPEG2TS	33
H263	34
SPEG	35
MPEG2VIDEO	36
AAC	37
WMA9STD	38
HEAAC	39
PCM_VOICE	40
PCM_AUDIO	41
AACLC	42
MP3	43
ADPCMA	49
AEC	50
X_LD	95
H264	96
D_GSM_HR	200
D_GSM_EFR	201
D_L8	202
D_RED	203
D_VDVI	204
D_BT656	220
D_H263_1998	221
D_MP1S	222
D_MP2P	223
D_BMPEG	224
MP4VIDEO	230
MP4AUDIO	237
VC1	238
JVC_ASF	255
D_AVI	256
MAX	257
AMR	258
MJPEG	259

Example

Request

```
{
  "MODULE": "MEDIASTREAMMODEL",
  "OPERATION": "REQUESTTALK",
  "SESSION": "21bc6f68-d2fd-4c3b-8aa4-c5fc1564ae51",
  "PARAMETER": {
    "CSRC": "INTERCOM",
    "SSRC": 65530,
    "SERIAL": 1,
    "STREAMNAME": "INTERCOMMEDIA",
    "MODE": 0,
    "CHANNEL": 1,
    "SOUNDMODE": 0,
    "CHANNELTOTAL": 32,
    "SAMPLINGRATE": 8000,
    "SAMPLINGFIGURE": 8,
    "AUDIOFORMAT": 19,
    "AUDIOSOURCE": 1,
    "AUDIOFRAMELEN": 320,
    "IPANDPORT": "58.60.231.218:5550"
  }
}
```

Response

```
{
  "MODULE": "MEDIASTREAMMODEL",
  "OPERATION": "REQUESTTALK",
  "SESSION": "21bc6f68-d2fd-4c3b-8aa4-c5fc1564ae51",
  "RESPONSE": {
    "STREAMNAME": "INTERCOMMEDIA",
    "SERIAL": 1,
    "CHANNEL": 1,
    "SOUNDMODE": 0,
    "CHANNELTOTAL": 1,
    "SAMPLINGRATE": 8000,
    "SAMPLINGFIGURE": 8,
    "AUDIOFORMAT": 19,
    "AUDIOSOURCE": 1,
    "AUDIOFRAMELEN": 320,
    "ERRORCODE": 0,
    "ERRORCAUSE": "SUCCESS"
  }
}
```

Intercom Control

Function Description

Intercom control

Protocol Introduction

MODULE

MEDIASTREAMMODEL

OPERATION

CONTROLTALK

PARAMETER

Attribute	Type	Value range	Description
CSRC	STRING	0~64 Bytes	Server identifier
SSRC	INTEGER	0~255	Channel number
STREAMNAME	STRING	1~32 Bytes	The name of media channel
CMD	INTEGER	0	Control media task operation. 0: stop
SERIAL	INTEGER		serial number

RESPONSE

Attribute	Type	Value range	Description
ERRORCODE	INTEGER		refer to the definition of error code
ERRORCAUSE	STRING	less than 100 characters	refer to the definition of error cause
STREAMNAME	STRING	1~32 Bytes	The name of media channel
SERIAL	INTEGER		The same as the request

Example Request

```
{
  "MODULE": "MEDIASTREAMMODEL",
  "OPERATION": "CONTROLTALK",
  "SESSION": "21bc6f68-d2fd-4c3b-8aa4-c5fc1564ae51",
  "PARAMETER": {
    "CSRC": "INTERCOM",
    "SSRC": 1,
    "STREAMNAME": "INTERCOMMEDIA",
    "CMD": 0
  }
}
```

Response

```
{
  "MODULE": "MEDIASTREAMMODEL",
  "OPERATION": "CONTROLTALK",
  "SESSION": "21bc6f68-d2fd-4c3b-8aa4-c5fc1564ae51",
  "RESPONSE": {
    "STREAMNAME": "INTERCOMMEDIA",
    "ERRORCODE": 0,
    "ERRORCAUSE": "SUCCESS"
  }
}
```

Error_Code

Error Code Table

error code number (hexadecimal)	error code number (decimal)	the description of error code
0X00000000	0	successful
0X00000001	1	failed to acquire the device
0X00000002	2	failed to verify the identity
0X00000003	3	authentication timeout
0X00000004	4	check S0 failed
0X00000005	5	username or password error
0X00000006	6	execution failed
0X00000007	7	protocol analysis failed
0X00000008	8	failed to connect to the media server
0X00000009	9	failed to register the media link
0X0000000A	10	failed to create the media transmission thread
0X0000000B	11	Currently, the protocol doesn't support the function
0X0000000C	12	memory allocation failed
0X0000000D	13	no matching results were found
0X0000000E	14	network issue
0X0000000F	15	failed to connect to the EMAIL server
0X00000010	16	timing failed
0X00000011	17	failed to go offline forcibly
0X00000012	18	no permissions
0X00000013	19	forced to go offline due to low permissions
0X00000014	20	failed to Add username
0X00000015	21	the same username
0X00000016	22	failed to edit user
0X00000017	23	failed to delete user
0X00000018	24	too many online users
0X00000019	25	task exists
0X0000001A	26	insufficient sources or task is full

0X0000001B	27	illegal channel
0X0000001C	28	uncoded
0X0000001D	29	main stream not supported
0X0000001E	30	sub-stream not supported
0X0000001F	31	mobile phone stream not supported
0X00000020	32	failed to start live preview
0X00000021	33	the file doesn't exist
0X00000022	34	failed to acquire the upgrade file path
0X00000023	35	failed to receive the upgrade file
0X00000024	36	fail to check the upgrade file
0X00000025	37	failed to open the upgrade file
0X00000026	38	failed to capture photos
0X00000027	39	failed to send data
0X00000028	40	no such task
0X00000029	41	unknown error
0X0000002A	42	parameter error
0X0000002B	43	time error
0X0000002C	44	server connection timeout
0X0000002D	45	other users are configuring
0X0000002E	46	too frequent operation
0X0000002F	47	testing
0X00000030	48	sender address error
0X00000031	49	recipient address error
0X00000032	50	server fill-in error
0X00000033	51	failed to read data from the device
0X00000034	52	failed to start remote sync playback
0X00000035	53	failed to end remote sync playback
0X00000036	54	failed to drag
0X00000037	55	too many upgrade users
0X00000038	56	the user has no permissions
0X00000039	57	the version number of upgrade file is the same
0X0000003A	58	the system is upgrading
0X0000003B	59	server port fill-in error

0X0000003C	60	WIFI module doesn't exist
0X0000003D	61	wWIFI switch is off
0X0000003E	62	linking
0X0000003F	63	user's MAC address is illegal
0X00000040	64	no permissions
0X00000041	65	media link exists
0X00000042	66	device serial number doesn't exist
0X00000043	67	audio coding failed
0X00000044	68	audio analysis failed
0X00000045	69	failed to start talkback
0X00000046	70	failed to download log
0X00000047	71	verifying
0X00000048	72	verification failed
0X00000049	73	successful verification
0X0000004A	74	acquiring valid IP
0X0000004B	75	failed to export parameters
0X0000004C	76	failed to import parameters
0X0000004D	77	parameter check failed
0X0000004E	78	failed to configure network parameters
0X0000004F	79	re-report to the server
0X00000050	80	failed to connect to the signaling server
0X00000051	81	processing command
0X00000052	82	no remote device
0X00000053	83	failed to log in the FTP server
0X00000054	84	upgrade package download is successful
0X00000055	85	no storage space or storage medium
0X00000056	86	disk is protected
0X00000057	87	the file is in the backup
0X00000058	88	task completed
0X00000059	89	task rejected
0X0000005A	90	USB upgrade interrupted
0X0000005B	91	remote playback data end
0X0000005C	92	the line is busy

0X0000005D	93	dialing failed
0X0000005E	94	not supported by the current network
0X0000005F	95	3G/4G modules start to upgrade
0X00000060	96	no deleted data
0X00000061	97	deletion is ready
0X00000062	98	deleting
0X00000064	100	user expired
0X00000065	101	time shift task has been deleted due to expiration
0X00000066	102	failed to receive operation & maintenance records
0X00000067	103	the file exists
0X00000068	104	failed to create file
0X00000069	105	USB undetected
0X0000006A	106	USB read/write error
0X0000006B	107	no corresponding file in the USB
0X0000006C	108	no hardware configuration errorCodeTable in the device
0X0000006E	110	failed to import hardware configuration errorCodeTable
0X0000006F	111	need to change the initial password
0X00000070	112	failed to save the new password
0X00000071	113	old and new passwords are the same
0X00000072	114	new and initial passwords are the same
0X00000073	115	the function is disabled
0X00000074	116	the task is in progress
0X00000075	117	download failed and no SD card in the passenger counter
0X00000076	118	no corresponding time period data in the SD card of passenger counter
0X00000077	119	failed to create routing
0X00000078	120	canceled task causes execution termination
0X00000079	121	cannot download video outside the Geo-fence
0X00000080	128	driver ID conflict
0X00000081	129	face matching not enabled
0X00000082	130	key verification failed
0X00000083	131	key type mismatch